

Disaster Alert Hotlines

I. NATIONAL LEVEL - GENERAL EMERGENCY & COORDINATION

National Emergency Hotline: 911

- *Scope:* Nationwide security, fire, and medical emergencies.

National Disaster Risk Reduction and Management Council (NDRRMC)

- **Trunklines:** (02) 8911-5061 to 65 (Local 100)
- **Operations Center (24/7):**
 - (02) 8911-1406
 - (02) 8912-2665
 - (02) 8912-5668
 - (02) 8911-1873

Philippine Red Cross (PRC)

- **Emergency Hotline: 143** (Free for Globe/Smart)
- **National Headquarters Trunkline:** (02) 8790-2300
- **Operations Center / Dispo:** (02) 8790-2300 loc. 604
- **Direct Emergency Lines:** (02) 8527-8385 to 95
- **Blood Center:** (02) 8527-0000

II. NATIONAL LEVEL - SECURITY & PUBLIC SAFETY

Philippine National Police (PNP)

- **Emergency Hotline: 117 or 911**
- **National Operation Center:** (02) 8722-0650
- **Text Hotline:** 0917-847-5757 (**Globe**)
- **Public Information Office:** (02) 8723-0401

Bureau of Fire Protection (BFP)

- **National Office Trunklines:**
 - (02) 8426-0219
 - (02) 8426-0246
 - (02) 8426-3812

Armed Forces of the Philippines (AFP)

- **Public Affairs Office:** (02) 8911-6001 loc. 6700
- **Civil Relations Service:** (02) 8911-6001 loc. 5640

III. NATIONAL LEVEL - SPECIALIZED HAZARD AGENCIES

PAGASA (Weather, Typhoon & Flood)

- **Trunkline:** (02) 8284-0800
- **Weather Forecasting Section:** (02) 8926-4258 / (02) 8927-1541
- **Aviation Weather (Aeronautical):** (02) 8832-3023

PHIVOLCS (Earthquake, Volcano & Tsunami)

- **Trunkline:** (02) 8426-1468 to 79
- **Seismology Division (Earthquake):** Locals 124 / 125
- **Volcano Monitoring Division:** Local 310

Philippine Coast Guard (PCG) - Maritime Search & Rescue

- **Action Center (Emergency):** (02) 8527-3877
- **Trunkline:** (02) 8527-8481 to 89
- **Mobile / Text Hotlines:**
 - 0917-724-3682 (**Globe**)
 - 0918-967-4697 (**Smart**)

Department of Health (DOH) - Health Emergencies

- **Health Emergency Management Bureau (HEMB):**
 - (02) 8711-1001
 - (02) 8711-1002
- **DOH Trunkline:** (02) 8651-7800

IV. NATIONAL LEVEL - SOCIAL WELFARE & TRANSPORT

Department of Social Welfare and Development (DSWD)

- **Disaster Response Management Bureau (DRMB):**
 - (02) 8856-3665
 - (02) 8852-8081
- **Text Hotline (Disaster):** 0918-912-2813 (**Smart**)
- **Agency Trunkline:** (02) 8931-8101 to 07

Department of Public Works and Highways (DPWH)

- **24/7 Hotline:** 165-02
- **Mobile Hotline:** 0961-684-7084 (**Smart/TNT**)
- **Bureau of Maintenance:** (02) 5304-3000

Department of Transportation (DOTr)

- **Action Center Hotline:** 7890
- **Land Transportation Office (LTO):** Text "LTOHELP" to 2600

Metro Manila Development Authority (MMDA)

- **Hotline: 136**
- **Flood Control:** (02) 8882-0925
- **Rescue Unit:** Local 337
- **Metrobase:** Local 255

Department of Agriculture (DA)

- **Hotline: 1381**
- **Trunkline:** (02) 8928-8741 to 64

V. EXPRESSWAY EMERGENCY HOTLINES

NLEX / SCTEX / NLEX Connector: Dial 1-35000

SLEX / Skyway / NAIAX / STAR Tollway:

- Landline: (02) 8824-2282
 - **Globe:** 0917-687-7539
 - Smart: 0999-886-0893
- CAVITEX (Manila-Cavite Expressway):
- Landline: (02) 8825-4004
 - Sun/Smart: 0942-822-8489
- CALAX (Cavite-Laguna Expressway): Dial 1-35000

VI. REGIONAL LEVEL (OFFICE OF CIVIL DEFENSE - OCD)

a. Luzon

National Capital Region (NCR)

- Landline: (02) 8421-1918 / (02) 8913-2786
- **Globe:** 0917-827-6325

Cordillera Administrative Region (CAR) - Baguio/Benguet Area

- Landline: (074) 304-2256 / (074) 619-0986
- **Globe:** 0916-507-0989

Region I (Ilocos Region)

- Landline: (072) 607-6528
- **Globe:** 0917-300-5096

Region II (Cagayan Valley)

- Landline: (078) 304-1630 / (078) 304-1631
- **Globe:** 0927-425-1954

Region III (Central Luzon)

- Landline: (045) 455-1526 / (045) 455-0033
- **Smart:** 0939-939-3051

Region IV-A (CALABARZON)

- Landline: (049) 531-7266 / (049) 834-4244
- **Smart:** 0908-889-8948

Region IV-B (MIMAROPA)

- Landline: (043) 723-4248 / (043) 702-9361
- **Globe:** 0917-533-0400

Region V (Bicol Region)

- Landline: (052) 742-1176
- **Globe:** 0917-574-7880
- **Smart:** 0928-505-3861

b. Visayas**Region VI (Western Visayas)**

- Landline: (033) 336-9353 / (033) 337-6671
- **Smart:** 0949-180-6115

Region VII (Central Visayas)

- Landline: (032) 416-5025 / (032) 253-8730
- **Globe:** 0917-704-1070 / 0917-947-5666
- **Smart:** 0949-471-0009

Region VIII (Eastern Visayas)

- Landline: (053) 323-7131
- **Globe:** 0917-842-7606 / 0917-700-1121

c. Mindanao**Region IX (Zamboanga Peninsula)**

- Landline: (062) 991-3450 / (062) 991-2365
- **Globe:** 0915-647-8884

Region X (Northern Mindanao)

- Landline: (088) 857-3907 / (088) 857-3988
- **Globe:** 0917-159-4486

Region XI (Davao Region)

- Landline: (082) 297-7915 / (082) 233-0295
- **Globe:** 0917-178-9711
- **Smart:** 0939-373-0049

Region XII (SOCCSKSARGEN)

- Landline: (083) 553-2994 / (083) 301-2994
- **Globe:** 0917-770-7771 / 0917-628-3720
- **Smart:** 0920-976-4001

Region XIII (CARAGA)

- Landline: (085) 342-8753 / (085) 815-6345
- **Smart:** 0947-946-8145

BARMM (Bangsamoro Autonomous Region)

- Landline: (064) 421-1669
- **Globe:** 0927-712-7466

VII. REGIONAL & PROVINCIAL HOTLINES (LUZON)

National Capital Region (NCR)

- **OCD NCR (Regional):** (02) 8421-1918 / 0917-827-6325 (Globe)
- **Quezon City:** 122 (Helpline) / 0947-885-9929 (Smart)
- **Manila:** (02) 8527-3087
- **Makati:** 168 / (02) 8236-5790
- **Pasig:** (02) 8643-0000 / 0908-899-3333 (Smart)
- **Marikina:** 161 / 0928-559-3341 (Smart)

Cordillera Administrative Region (CAR)

- **OCD CAR (Regional):** (074) 619-0986 / 0916-507-0989 (Globe)
- **Abra:** (074) 752-8118 / 0917-538-6767 (Globe)
- **Apayao:** 0917-519-3757 (Globe)
- **Baguio City:** 911 / 0927-628-0498 (Globe) / 0998-598-7739 (Smart)
- **Benguet:** (074) 422-2307 / 0998-598-7739 (Smart)
- **Ifugao:** 0917-160-2251 (Globe)
- **Kalinga:** 0917-530-2678 (Globe)
- **Mountain Province:** 0917-530-3306 (Globe)

Region I (Ilocos Region)

- **OCD Region I:** (072) 607-6528 / 0917-300-5096 (Globe)
- **Ilocos Norte:** 911 / 0917-109-9911 (Globe)

- **Ilocos Sur:** (077) 722-2776 / 0917-568-1872 (Globe)
- **La Union:** 911 / 0998-561-1519 (Smart)
- **Pangasinan:** 0917-150-5754 (Globe) / 0918-934-5754 (Smart)

Region II (Cagayan Valley)

- **OCD Region II:** (078) 304-1630 / 0917-544-4970 (Globe)
- **Batanes:** 0929-363-5545 (Smart)
- **Cagayan:** 0917-581-4333 (Globe) / 0927-181-9424 (Globe)
- **Isabela:** (078) 323-0416 / 0917-852-6698 (Globe / DART Rescue)
- **Nueva Vizcaya:** (078) 332-3000 / 0917-104-1573 (Globe)
- **Quirino:** 0917-500-6638 (Globe)

Region III (Central Luzon)

- **OCD Region III:** (045) 455-1526 / 0939-939-3051 (Smart)
- **Aurora:** 0921-376-7429 (Smart) / (042) 724-3221
- **Bataan:** 911 / 0909-507-7158 (Smart)
- **Bulacan:** 911 / (044) 791-0566 / 0905-333-3319 (Globe)
- **Nueva Ecija:** 0916-362-2365 (Globe) / 0921-946-6959 (Smart)
- **Pampanga:** (045) 435-2533 / 0917-706-6730 (Globe)
- **Tarlac:** (045) 925-5126 / 911
- **Zambales:** (047) 602-1866 / 0920-958-7093 (Smart)

Region IV-A (CALABARZON)

- **OCD IV-A:** (049) 531-7266 / 0908-889-8948 (Smart)
- **Batangas:** (043) 723-9350 / 0917-506-2866 (Globe)
- **Cavite:** (046) 471-0629 / 0998-849-9635 (Smart)
- **Laguna:** (049) 501-4373 / 911
- **Quezon:** (042) 710-3933 / 0946-274-8706 (Smart)
- **Rizal:** (02) 8571-4375 / 0961-869-5888 (Smart/TNT)

Region IV-B (MIMAROPA)

- **OCD IV-B:** (043) 723-4248 / 0917-533-0400 (Globe)
- **Marinduque:** (042) 332-0834 / 0929-708-0711 (Smart)
- **Occidental Mindoro:** 911 / 0909-965-3531 (Smart - Magsaysay MDRRMO)
- **Oriental Mindoro:** (043) 288-7935 / 0917-808-0698 (Globe)
- **Palawan:** 0917-775-6640 (Globe)
- **Romblon:** 911 / 0998-546-0269 (Smart)

Region V (Bicol Region)

- **OCD Region V:** (052) 742-1176 / 0917-574-7880 (Globe)
- **Albay:** (052) 742-6766 / 0926-828-4392 (Globe)
- **Camarines Norte:** 0917-583-9147 (Globe) / 0998-561-5388 (Smart)
- **Camarines Sur:** 911 / 0917-159-9892 (Globe - Police)

- **Catanduanes:** 0929-961-1026 (Smart - Bato MDRRMO) / 0939-831-4518 (PDRRMO)
- **Masbate:** (056) 582-1254 / 0917-504-5309 (Globe)
- **Sorsogon:** 911 / 0919-911-1111 (Smart)

VIII. REGIONAL & PROVINCIAL HOTLINES (VISAYAS)

Region VI (Western Visayas)

- **OCD Region VI:** (033) 336-9353 / 0949-180-6115 (Smart)
- **Aklan:** (036) 268-5333 / 0939-530-7666 (Smart)
- **Antique:** (036) 540-7136 / 0917-335-7631 (Globe)
- **Capiz:** (036) 621-0042 / 0912-472-2669 (Smart/TNT - Roxas CDRRMO)
- **Guimaras:** (033) 581-2184 / 0961-684-7084 (Smart)
- **Iloilo:** (033) 328-7920 / 0998-954-5097 (Smart)
- **Negros Occidental:** (034) 432-3871 / 0930-243-4706 (Smart)

Region VII (Central Visayas)

- **OCD Region VII:** (032) 253-8730 / 0917-947-5666 (Globe)
- **Bohol:** 911 / 117 (TaRSIER 117 Rescue)
- **Cebu Province:** (032) 888-2328 / 0917-813-3611 (Globe)
- **Negros Oriental:** 911 / 0965-907-6449 (Globe)
- **Siquijor:** 0953-209-9165 (Globe) / 0961-617-5474 (Smart)

Region VIII (Eastern Visayas)

- **OCD Region VIII:** (053) 323-7131 / 0917-700-1121 (Globe)
- **Biliran:** (053) 500-0091 / 0915-641-9144 (Globe)
- **Eastern Samar:** (055) 261-2414
- **Leyte:** (053) 323-3018 / 0968-361-7021 (Smart)
- **Northern Samar:** 0967-339-5807 (Globe) / 0947-100-7033 (Smart)
- **Samar (Western):** (055) 251-2196
- **Southern Leyte:** (053) 570-7108

IX. REGIONAL & PROVINCIAL HOTLINES (MINDANAO)

Region IX (Zamboanga Peninsula)

- **OCD Region IX:** (062) 991-3450 / 0915-647-8884 (Globe)
- **Zamboanga del Norte:** (065) 212-9667
- **Zamboanga del Sur:** (062) 214-3789
- **Zamboanga Sibugay:** (062) 333-5507

Region X (Northern Mindanao)

- **OCD Region X:** (088) 857-3907 / 0917-159-4486 (Globe)

- **Bukidnon:** (088) 813-1262 / 0917-631-0004 (Globe)
- **Camiguin:** (088) 387-1018
- **Lanao del Norte:** (063) 341-5241
- **Misamis Occidental:** 0938-002-5372 (Clarín MDRRMO)
- **Misamis Oriental:** (088) 856-5059

Region XI (Davao Region)

- **OCD Region XI:** (082) 297-7915 / 0939-373-0049 (Smart)
- **Davao de Oro:** (084) 376-1066
- **Davao del Norte:** (084) 400-2167 / 911
- **Davao del Sur:** 0927-225-4572 (Globe)
- **Davao Occidental:** 0997-208-9834 (Globe)
- **Davao Oriental:** 0917-630-3224 (Globe)

Region XII (SOCCSKSARGEN)

- **OCD Region XII:** (083) 553-2994 / 0917-628-3720 (Globe)
- **Cotabato:** (064) 521-8080
- **Sarangani:** (083) 508-2047 / 0927-676-8676 (Globe - Malungon MDRRMO)
- **South Cotabato:** (083) 419-0000 / (083) 228-3754
- **Sultan Kudarat:** (064) 200-3153

Region XIII (CARAGA)

- **OCD CARAGA:** (085) 342-8753 / 0947-946-8145 (Smart)
- **Agusan del Norte:** 0948-054-5114 (Smart) / 0985-868-2872 (Talk N Text)
- **Agusan del Sur:** 0948-679-8454 (Smart)
- **Dinagat Islands:** 0963-537-3604 (Smart - MDRRMO)
- **Surigao del Norte:** (086) 826-0144
- **Surigao del Sur:** (086) 211-3258

BARMM (Bangsamoro Autonomous Region)

- **BARMM READi:** (064) 421-1669 / 0927-712-7466 (Globe)
- **Basilan:** 0927-631-7768 (Globe)
- **Lanao del Sur:** 0907-955-1205 (Smart) / 0906-532-9895 (Globe)
- **Maguindanao:** (064) 489-0903
- **Sulu:** (085) 341-8911
- **Tawi-Tawi:** 911

X. CITY LEVEL EMERGENCY HOTLINES (LUZON)

NATIONAL CAPITAL REGION (NCR)

Note: Area Code for all is (02)

1. **Caloocan City :**
 - Caloocan City DRRMO / Command Center: 888-25664 / 8310-6972

- PNP (Police): 8364-8064
- BFP (Fire): 8362-4044
- 2. **Las Piñas City :**
 - LP Disaster Response Office: 8552-7694 / 8255-6677
 - PNP: 8808-7936
 - BFP: 8282-3788
- 3. **Makati City :**
 - Makati Command Control (C3): 168 (Shortcode) / 8236-5790
 - PNP: 8887-1798
 - BFP: 8818-5150
- 4. **Malabon City :**
 - Malabon Command Center: 8921-6009 / 8921-6029
 - PNP: 8281-1138
 - BFP: 8281-0854
- 5. **Mandaluyong City :**
 - Mandaluyong DRRMO / C3: 8533-2225 / 8534-2220
 - PNP: 8532-2145
 - BFP: 8532-2402
- 6. **Manila (City of Manila) :**
 - Manila DRRMO (MDRRMO): 8527-5174 / 871-5811
 - Manila Emergency Response: 8925-3333
 - BFP (Manila Main): 8527-3600
- 7. **Marikina City :**
 - Marikina Rescue 161: 161 (Shortcode) / 8646-2436
 - PNP: 8942-2630
 - BFP: 8681-0202
- 8. **Muntinlupa City :**
 - Muntinlupa Rescue: 137-175 (Shortcode) / 8925-4351
 - PNP: 8862-2611
 - BFP: 8842-2266
- 9. **Navotas City :**
 - Navotas TXT JRT (Action Center): 8281-1111 / 8281-8531
 - PNP: 8281-9099
 - BFP: 8281-0854
- 10. **Parañaque City :**
 - Parañaque DRRMO: 8820-7783 / 911 (Local)
 - PNP: 8829-2114
 - BFP: 8826-9131
- 11. **Pasay City :**
 - Pasay DRRMO: 8551-7777 / 8833-8512
 - PNP: 8831-1544
 - BFP: 8844-2120
- 12. **Pasig City :**
 - Pasig City DRRMO: 8643-0000
 - Pasig PNP: 8477-7953
 - Pasig BFP: 8641-2815
- 13. **Quezon City :**

- QC Helpline: 122 (Shortcode)
- QC DRRMO: 8927-5914
- PNP (QCPD): 8921-7474
- BFP: 8928-8363

14. San Juan City :

- San Juan DRRMO: 7238-4333 (loc 100) / 137-135
- PNP: 8724-2515
- BFP: 8725-2079

15. Taguig City :

- Taguig Command Center: 8789-3200
- PNP: 8642-3582
- BFP: 8837-0740

16. Valenzuela City :

- Valenzuela Rescue (VCDRRMO): 8352-5000 / 8292-1405
- PNP: 8352-4000
- BFP: 8292-0235

CORDILLERA ADMINISTRATIVE REGION (CAR)

Area Code: (074)

17. Baguio City :

- Baguio City DRRMO / Smart City Command: 442-1900 / 442-1905 / 911
- PNP: 442-1211
- BFP: 442-2222

18. Tabuk City :

- Tabuk City DRRMO: 0917-146-5332 (Mobile)
- PNP: 0908-193-6744
- Kalinga Provincial Hospital: (074) 627-5896

REGION I (ILOCOS REGION)

Area Code: (077) for Ilocos, (075) for Pangasinan

19. Alaminos City :

- Alaminos DRRMO / Action Center: (075) 552-7777 / 0908-869-7777
- PNP: (075) 552-7107

20. Batac City :

- Batac DRRMO: (077) 792-2144
- PNP: (077) 792-3168

21. Candon City :

- Candon DRRMO: (077) 742-5369
- PNP: (077) 742-6003

22. Dagupan City :

- Dagupan CDRRMC: (075) 522-8302 / 0947-521-8270
- PNP: (075) 515-4309

23. Laoag City :

- Laoag City DRRMO: (077) 772-0001
- PNP: (077) 772-0201
- Gov. Roque Ablan Sr. Memorial Hospital: (077) 772-1875
- 24. San Carlos City (Pangasinan) :**
 - San Carlos DRRMO: (075) 955-5353
 - PNP: (075) 532-2303
- 25. San Fernando City (La Union) :**
 - San Fernando CDRRMO: (072) 687-8100 (loc 133) / 0995-658-4509
 - PNP: (072) 607-2856
- 26. Urdaneta City :**
 - Urdaneta DRRMO: (075) 656-1724 / 0998-550-9907
 - PNP: (075) 568-2319
- 27. Vigan City :**
 - Vigan City DRRMO: (077) 722-2683
 - Vigan PNP: (077) 722-2639

REGION II (CAGAYAN VALLEY)

Area Code: (078)

- 28. Cauayan City :**
 - Cauayan City Rescue 922: (078) 652-2000 / 0927-425-6031
 - PNP: (078) 652-2015
- 29. Ilagan City :**
 - Ilagan City DRRMO / Rescue 1124: (078) 624-0320 / 0917-862-2432
 - PNP: (078) 622-2121
- 30. Santiago City :**
 - Santiago City DRRMO: (078) 305-1888
 - PNP: (078) 305-2748
- 31. Tuguegarao City :**
 - Tuguegarao CDRRMO / Command Center: (078) 304-1114 / 0917-578-1029
 - PNP: (078) 844-1629

REGION III (CENTRAL LUZON)

Area Codes vary: (044) Bulacan/NE, (045) Pampanga/Tarlac, (047) Zambales

- 32. Angeles City :**
 - Angeles City DRRMO (ACDRRMO): (045) 322-7796 / 893-2212
 - PNP: (045) 322-7714
- 33. Balanga City :**
 - Balanga CDRRMO: (047) 237-0687
 - PNP: (047) 237-2444
- 34. Cabanatuan City :**
 - Cabanatuan DRRMO: (044) 940-0648 / 0919-456-6211
 - PNP: (044) 463-1496

35. Gapan City :

- Gapan DRRMO: 0942-012-3252 (Mobile) / (044) 486-0513
- PNP: (044) 486-0209

36. Mabalacat City :

- Mabalacat DRRMO / CDRRMC: (045) 331-5000 / 0998-583-1768
- PNP: (045) 331-2092

37. Malolos City :

- Malolos CDRRMO: (044) 796-2793 / 0928-273-5183
- PNP: (044) 791-0536

38. Meycauayan City :

- Meycauayan DRRMO: (044) 769-6379 / 0925-709-6611
- PNP: (044) 840-9759

39. Muñoz (Science City of Muñoz) :

- Muñoz DRRMO: (044) 456-0100
- PNP: (044) 456-0164

40. Olongapo City :

- Olongapo DRRMO: (047) 222-2626 / 0929-166-3023
- PNP: (047) 222-2571

41. Palayan City :

- Palayan City DRRMO: (044) 958-5004 / 0998-964-4299
- PNP: (044) 940-5975

42. San Fernando City (Pampanga) :

- San Fernando Rescue (SAFRU): (045) 961-4568 / 0917-539-7568
- PNP: (045) 961-2453

43. San Jose City (Nueva Ecija) :

- San Jose DRRMO: (044) 511-1976
- PNP: (044) 511-1634

44. San Jose del Monte (CSJDM) :

- CSJDM DRRMO: (044) 815-5858 / 0905-277-2244
- PNP: (044) 691-7557

45. Tarlac City :

- Tarlac City DRRMO: (045) 611-3000
- PNP: (045) 982-1678

46. Baliwag City (New City) :

- Baliwag DRRMO: (044) 798-1582
- PNP: (044) 766-2151

REGION IV-A (CALABARZON)

Area Codes: (046) Cavite, (049) Laguna, (043) Batangas, (042) Quezon/Aurora, (02) Rizal

47. Antipolo City :

- Antipolo DRRMO (OPCEN): (02) 8689-4564 / 8689-4573
- PNP: (02) 8697-2489

48. Bacoor City :

- Bacoor DRRMO: (046) 417-0707 / 0915-463-5481

- PNP: (046) 417-6366
- 49. **Batangas City :**
 - Batangas City CDRRMO: (043) 702-3902 / 980-0385
 - PNP: (043) 723-2030
- 50. **Biñan City :**
 - Biñan Command Center: (049) 511-9111 / 0917-543-8547
 - PNP: (049) 511-4144
- 51. **Cabuyao City :**
 - Cabuyao DRRMO: (049) 531-3141 / 0920-842-8453
 - PNP: (049) 531-2137
- 52. **Calamba City :**
 - Calamba DRRMO: (049) 545-5162
 - PNP: (049) 545-6789
- 53. **Cavite City :**
 - Cavite City DRRMO: (046) 431-0164
 - PNP: (046) 431-0306
- 54. **Dasmariñas City :**
 - Dasmariñas CDRRMO: (046) 416-0000 / 0905-472-8800
 - PNP: (046) 416-0268
- 55. **General Trias City :**
 - GenTri DRRMO: (046) 509-5053 / 0917-543-2678
 - PNP: (046) 437-7505
- 56. **Imus City :**
 - Imus CDRRMO: (046) 471-2294 / 0917-865-9831
 - PNP: (046) 471-0428
- 57. **Lipa City :**
 - Lipa CDRRMO / Rescue 911: (043) 756-1212
 - PNP: (043) 756-2592
- 58. **Lucena City :**
 - Lucena DRRMO: (042) 710-3844 / 0939-906-8969
 - PNP: (042) 373-0498
- 59. **San Pablo City :**
 - San Pablo CDRRMO: (049) 800-0493
 - PNP: (049) 562-6500
- 60. **San Pedro City :**
 - San Pedro DRRMO: (02) 8808-2020 (loc 108) / 0998-592-2736
 - PNP: (02) 8808-1856
- 61. **Santa Rosa City :**
 - Santa Rosa DRRMO: (049) 534-1906 / 0917-543-0518
 - PNP: (049) 534-1383
- 62. **Santo Tomas City :**
 - Sto. Tomas DRRMO: (043) 784-8022
 - PNP: (043) 778-4333
- 63. **Tagaytay City :**
 - Tagaytay DRRMO: (046) 483-0546
 - PNP: (046) 413-1282
- 64. **Tanauan City :**

- Tanauan CDRMO / Rescue: (043) 778-5095 / 0939-923-9686
- PNP: (043) 778-1123

65. Tayabas City :

- Tayabas DRRMO: (042) 793-3563
- PNP: (042) 793-3243

66. Trece Martires City :

- Trece Martires DRRMO: (046) 419-0164
- PNP: (046) 419-0268

67. Carmona City (New City) :

- Carmona DRRMO: (046) 430-1596
- PNP: (046) 430-1707

REGION IV-B (MIMAROPA)

Area Code: (048) Palawan, (043) Mindoro

68. Calapan City :

- Calapan CDRMO: (043) 288-7111 / 0917-807-6469
- PNP: (043) 288-2525

69. Puerto Princesa City :

- Puerto Princesa 911 / CDRMO: (048) 434-9425 / 911
- PNP: (048) 434-9888

REGION V (BICOL REGION)

Area Code: (052) Albay, (054) CamSur, (056) Sorsogon

70. Iriga City :

- Iriga DRRMO: (054) 299-2508
- PNP: (054) 299-2325

71. Legazpi City :

- Legazpi 911 Command Center: (052) 742-0313 / 911
- PNP: (052) 481-0610

72. Ligao City :

- Ligao CDRMO: (052) 485-1116
- PNP: (052) 485-1127

73. Masbate City :

- Masbate City DRRMO: (056) 333-5844 / 0998-550-9844
- PNP: (056) 333-2256

74. Naga City :

- Naga City COPC / CDRMO: (054) 811-2562 / 0908-886-4560
- PNP: (054) 473-3537
- Bicol Medical Center: (054) 472-6125

XI. CITY LEVEL EMERGENCY HOTLINES (VISAYAS)

REGION VI (WESTERN VISAYAS)

Area Codes: (033) Iloilo, (034) Negros Occ., (036) Capiz/Aklan/Antique

1. **Bacolod City:**
 - DRRMO / 911: (034) 432-3871 / 911
 - PNP: (034) 433-0300
2. **Bago City:**
 - DRRMO: (034) 461-1258
 - PNP: (034) 461-0169
3. **Cadiz City:**
 - DRRMO: (034) 493-0211 / 0929-322-2970
 - PNP: (034) 493-0369
4. **Escalante City:**
 - DRRMO: (034) 454-0238
 - PNP: (034) 454-0370
5. **Himamaylan City:**
 - DRRMO: (034) 388-3645 / 0939-915-0824
 - PNP: (034) 388-3607
6. **Iloilo City:**
 - ICER (Emergency): (033) 333-2333 / 753-3333
 - PNP (ICPO): (033) 337-0400
7. **Kabankalan City:**
 - DRRMO / 911: (034) 471-2287 / 0908-185-9966
 - PNP: (034) 471-2234
8. **La Carlota City:**
 - DRRMO: (034) 460-2580
 - PNP: (034) 460-2350
9. **Passi City:**
 - DRRMO: (033) 311-5369
 - PNP: (033) 311-5026
10. **Roxas City:**
 - DRRMO (CERT): (036) 621-0042 / 522-2630
 - PNP: (036) 621-0329
11. **Sagay City:**
 - DRRMO: (034) 488-0101 / 722-0900
 - PNP: (034) 722-4217
12. **San Carlos City (Negros Occ.):**
 - DRRMO: (034) 312-5678 / 729-3330
 - PNP: (034) 312-6134
13. **Silay City:**
 - DRRMO: (034) 495-5020
 - PNP: (034) 495-4666
14. **Sipalay City:**
 - DRRMO: (034) 473-2101 / 0946-402-9590
 - PNP: (034) 473-2066
15. **Talisay City (Negros Occ.):**
 - DRRMO: (034) 495-4428

- PNP: (034) 495-4144
- 16. Victorias City:**
 - DRRMO: (034) 399-3449 / 399-2996
 - PNP: (034) 399-3353

REGION VII (CENTRAL VISAYAS)

Area Codes: (032) Cebu, (035) Negros Or., (038) Bohol

- 17. Bais City:**
 - DRRMO: (035) 402-8113
 - PNP: (035) 402-8105
- 18. Bayawan City:**
 - DRRMO: (035) 531-0294 / 430-0196
 - PNP: (035) 531-0012
- 19. Bogu City:**
 - DRRMO: (032) 260-5188 / 0917-310-9831
 - PNP: (032) 434-8077
- 20. Canlaon City:**
 - DRRMO: (035) 414-0003
 - PNP: (035) 414-0005
- 21. Carcar City:**
 - DRRMO: (032) 487-9777 / 231-5085
 - PNP: (032) 487-9076
- 22. Cebu City:**
 - CCDRRMO (Command Center): (032) 262-1424 / 166
 - PNP (CCPO): (032) 231-1823
- 23. Danao City:**
 - DRRMO: (032) 200-3333
 - PNP: (032) 200-3392
- 24. Dumaguete City:**
 - DRRMO / One Rescue: (035) 225-3850 / 422-9721
 - PNP: (035) 225-1768
- 25. Guihulngan City:**
 - DRRMO: 0917-805-4654 / 0995-188-6933
 - PNP: (035) 410-4100
- 26. Lapu-Lapu City:**
 - DRRMO: (032) 341-3771 / 340-2124
 - PNP: (032) 341-1311
- 27. Mandaue City:**
 - Bantay Mandaue (DRRMO): (032) 383-1658 / 344-3356
 - PNP: (032) 344-1200
- 28. Naga City (Cebu):**
 - DRRMO: (032) 272-6655 / 489-4911
 - PNP: (032) 272-6660
- 29. Tagbilaran City:**
 - DRRMO: (038) 411-2911

- PNP: (038) 411-3083
- 30. Talisay City (Cebu):**
 - DRRMO: (032) 273-3628 / 516-2720
 - PNP: (032) 272-4666
- 31. Tanjay City:**
 - DRRMO: (035) 415-9400
 - PNP: (035) 415-8461
- 32. Toledo City:**
 - DRRMO: (032) 322-5788 / 467-9677
 - PNP: (032) 322-5776

REGION VIII (EASTERN VISAYAS)

Area Codes: (053) Leyte, (055) Samar

- 33. Baybay City:**
 - DRRMO: (053) 563-7107
 - PNP: (053) 335-2015
- 34. Borongan City:**
 - DRRMO: 0917-817-4958 (Globe) / 0918-959-1647 (Smart)
 - PNP: (055) 560-9117
- 35. Calbayog City:**
 - DRRMO: (055) 209-3298
 - PNP: (055) 209-1237
- 36. Catbalogan City:**
 - DRRMO: (055) 543-9889 / 0917-826-6268
 - PNP: (055) 251-2166
- 37. Maasin City:**
 - DRRMO: (053) 570-9679 / 570-8012
 - PNP: (053) 381-2106
- 38. Ormoc City:**
 - DRRMO / 911: (053) 255-9467 / 255-2454
 - PNP: (053) 255-2253
- 39. Tacloban City:**
 - TACRU (Tacloban Rescue): (053) 888-0911 / 832-0744
 - PNP: (053) 321-3011

XII. CITY LEVEL EMERGENCY HOTLINES (MINDANAO)

Comprehensive Directory of Local Command Centers for all 33 Cities in Mindanao regions.

REGION IX (ZAMBOANGA PENINSULA)

Area Codes: (062) Zamboanga/Basilan, (065) Zamboanga Norte, (062/063) Zamboanga Sur

- 1. Dapitan City:**
 - DRRMO (CDR): (065) 213-6439 / 908-8255
 - PNP: (065) 213-6412

2. **Dipolog City:**
 - DRRMO / Rescue 117: (065) 212-3456 / 908-8250
 - PNP: (065) 212-2309
3. **Isabela City (Basilan):**
 - DRRMO: (062) 200-3000
 - PNP: (062) 200-7634
4. **Pagadian City:**
 - DRRMO: (062) 215-2652 / 0948-660-3944
 - PNP: (062) 215-2638
5. **Zamboanga City:**
 - Command Center: 911 / (062) 992-3580
 - PNP (ZCPO): (062) 991-5656

REGION X (NORTHERN MINDANAO)

Area Codes: (088) Misamis Or./Bukidnon, (063) Lanao Norte, (088) Misamis Occ.

6. **Cagayan de Oro City:**
 - Oro Rescue / 911: (088) 857-3988 / 911
 - PNP (COCPO): (088) 853-7119
7. **El Salvador City:**
 - DRRMO: 0917-328-9111 / (088) 555-0553
 - PNP: (088) 555-0377
8. **Gingoog City:**
 - DRRMO: (088) 861-5079 / 427-4646
 - PNP: (088) 861-1393
9. **Iligan City:**
 - ICDRRMO: (063) 221-5071 / 161 (Smart/PLDT)
 - PNP: (063) 221-6699
10. **Malaybalay City:**
 - DRRMO: (088) 813-4318 / 0917-722-1925
 - PNP: (088) 813-3950
11. **Oroquieta City:**
 - DRRMO: (088) 531-1801
 - PNP: (088) 531-1311
12. **Ozamiz City:**
 - DRRMO: (088) 521-3311
 - PNP: (088) 521-0268
13. **Tangub City:**
 - DRRMO: (088) 354-1594
 - PNP: (088) 545-0062
14. **Valencia City:**
 - DRRMO: (088) 828-3696 / 0917-327-0402
 - PNP: (088) 828-2175

REGION XI (DAVAO REGION)

Area Codes: (082) Davao City/Sur, (084) Davao Norte/Oro, (087) Davao Oriental

15. Davao City:

- Central 911: 911 (Landline/Mobile)
- PNP (DCPO): (082) 224-1313

16. Digos City:

- DRRMO: (082) 553-8380 / 0928-356-8313
- PNP: (082) 553-2287

17. Island Garden City of Samal (IGaCoS):

- DRRMO: 0930-141-8742 / (082) 301-6532
- PNP: (082) 301-6380

18. Mati City:

- DRRMO: (087) 388-4332 / 0920-557-0744
- PNP: (087) 388-3482

19. Panabo City:

- DRRMO: (084) 628-5285 / 822-1662
- PNP: (084) 628-5437

20. Tagum City:

- Tagum 911: (084) 216-9111
- PNP: (084) 655-6372

REGION XII (SOCCSKSARGEN)

Area Codes: (083) South Cotabato/Sarangani, (064) Sultan Kudarat/Cotabato

21. General Santos City:

- CDRRMO: (083) 552-8484 / 552-3939
- PNP: (083) 553-3554

22. Kidapawan City:

- Call 911: (064) 577-4933 / 911
- PNP: (064) 577-1558

23. Koronadal City:

- DRRMO (Action Center): (083) 228-5369
- PNP: (083) 228-3233

24. Tacurong City:

- DRRMO: (064) 200-6644 / 0929-224-4061
- PNP: (064) 200-3333

REGION XIII (CARAGA)

Area Codes: (085) Agusan Norte/Sur, (086) Surigao Norte/Sur

25. Bayugan City:

- DRRMO: (085) 343-6775
- PNP: (085) 343-6111

26. Bislig City:

- DRRMO: (086) 853-2007

- PNP: (086) 853-3151
- 27. Butuan City:**
 - Rescue 111: (085) 815-6111 / 111 (Crurtel)
 - PNP (BCPO): (085) 342-3023
- 28. Cabadbaran City:**
 - DRRMO: (085) 818-5544 / 0912-888-8888
 - PNP: (085) 343-1111
- 29. Surigao City:**
 - DRRMO: (086) 826-0000 / 0909-511-1647
 - PNP: (086) 826-4444
- 30. Tandag City:**
 - DRRMO: (086) 211-5060
 - PNP: (086) 211-3069

BANGSAMORO AUTONOMOUS REGION (BARMM)

Area Codes: (064) Cotabato/Maguindanao, (063) Lanao Sur

- 31. Cotabato City:**
 - DRRMO / Rescue: (064) 421-2195 / 0917-683-1123
 - PNP (Station 1): (064) 421-2287
- 32. Lamitan City:**
 - DRRMO: 0975-403-9993
 - PNP: (062) 200-3663
- 33. Marawi City:**
 - DRRMO: (063) 876-0007 / 0917-709-6611
 - PNP: (063) 352-0992

EVACUATION SITES

DISCLAIMER:

A. The 90% Variable Rule Over 90% of evacuation centers in the Philippines are **temporary and dual-use facilities** (specifically Public Schools, Barangay Covered Courts, and Multi-Purpose Halls). These are not permanent evacuation centers. They are activated only upon the directive of the Local Government Unit (LGU).

B. Hazard Specificity (Conflict Logic) A facility listed as an evacuation site is not safe for all disaster types:

- **Flood Scenario:** Schools are often used (2nd/3rd floors), but single-story structures may be dangerous.
- **Earthquake Scenario:** Schools and covered courts may be structurally compromised. Open fields and stadiums are prioritized.
- **Volcanic Eruption:** Sites must be checked against the specific danger zone radius (e.g., Taal 14km danger zone).

C. User Verification Protocol Unless the site is listed below as a "Permanent/Mega Hub," the AI must append the following warning to any location suggestion:

*"Please verify availability with your local Barangay Captain or MDRRMO.
Evacuation sites change based on the specific hazard (e.g., a school used for floods may not be safe for earthquakes)."*

REGIONAL DIRECTORY OF PERMANENT & MAJOR EVACUATION SITES

I. NATIONAL CAPITAL REGION (NCR)

Priority: High-density population centers. Designated by Sector.

- **Manila (City of):**
 - Delpan Evacuation Center (Binondo) – *Permanent Structure*
 - Rizal Memorial Sports Complex (Malate)
 - Baseco Evacuation Center (Port Area)
- **Quezon City:**
 - Quezon Memorial Circle – *Designated Open-Air/Tent City*
 - Amoranto Sports Complex
 - UP Diliman Sunken Garden – *Designated Open-Air Triage*
- **Marikina City (Flood Focus):**
 - Marikina Sports Center (Main Hub)
 - Sto. Niño Elementary School
 - Malanday Elementary School
- **Pasig City:**
 - PhilSports Arena (ULTRA) – *Mega Ligas Center*
- **Pasay City:**

- Cuneta Astrodome
- Villamor Air Base Golf Course – *Open-Air*
- **Taguig City:**
 - Taguig City University Gym
- **Caloocan/Valenzuela:**
 - Valenzuela Astrodome
- **Muntinlupa:**
 - Alabang Country Club – *Open-Air*

II. CORDILLERA ADMINISTRATIVE REGION (CAR)

Priority: Landslide and Earthquake safety zones.

- **Baguio City:**
 - Baguio Convention Center (Main Hub)
 - PFVR Gym (Military Cut-off)
- **Benguet:**
 - Benguet Sports Complex (Wangal, La Trinidad)
- **Abra:**
 - Abra Sports Complex (Bangued)
- **Kalinga:**
 - Kalinga Sports Center (Tabuk City)

III. REGION 1 (ILOCOS REGION)

- **Ilocos Norte:**
 - Ilocos Norte Centennial Arena (Laoag City)
 - Pagudpud Cultural and Sports Complex
- **Ilocos Sur:**
 - Quirino Stadium (Vigan City)
 - Candon City Arena
- **La Union:**
 - La Union Convention Center (San Fernando City)
 - Agoo Civic Center
- **Pangasinan:**
 - Pangasinan Training and Development Center (Lingayen)
 - Dagupan City People's Astrodome
 - Urdaneta Cultural and Sports Center

IV. REGION 2 (CAGAYAN VALLEY)

- **Cagayan:**
 - Cagayan Sports Complex (Tuguegarao City)
 - People's Gymnasium (Tuguegarao City)
 - Lal-lo Evacuation Center – *Permanent DPWH Built*
- **Isabela:**
 - Isabela Provincial Capitol Amphitheater (Ilagan City)

- F.L. Dy Coliseum (Cauayan City)
- **Nueva Vizcaya:**
 - Nueva Vizcaya Convention Center (Bayombong)

V. REGION 3 (CENTRAL LUZON)

Priority: Large-scale "Mega" centers for mass displacement.

- **Pampanga:**
 - ASEAN Convention Center (Clark Freeport Zone) – *Mega Hub*
 - Bren Z. Guiao Convention Center (San Fernando City)
 - Lubao Evacuation Center (Brgy. Sta. Catalina) – *Permanent DPWH Built*
 - Mexico Evacuation Center (Brgy. San Jose Matulid) – *Permanent DPWH Built*
- **Tarlac:**
 - New Clark City Government Center & Stadium (Capas) – *National Back-up Hub*
- **Bulacan:**
 - Philippine Arena Complex (Bocaue) – *Designated Mega Tent Area*
 - Bulacan Sports Complex (Malolos City)
- **Nueva Ecija:**
 - Nueva Ecija Convention Center (Palayan City)
- **Zambales:**
 - Olongapo City Convention Center
 - Zambales Sports Complex (Iba)

VI. REGION 4-A (CALABARZON)

Priority: Taal Volcano Eruption designated zones and Flood Control.

- **Batangas:**
 - Batangas Provincial Sports Complex (Batangas City)
 - Dream Zone Capitol Compound (Batangas City)
 - Polytechnic University of the Philippines (PUP) Gymnasium (Sto. Tomas)
 - Taal Evacuation Center (Brgy. Cawit) – *Permanent DPWH Built*
- **Cavite:**
 - Tagaytay International Convention Center
 - Brgy. Calibuyo Evacuation Center (Tanza) – *Permanent DPWH Built*
- **Laguna:**
 - Alonte Sports Arena (Biñan City) – *Mega Ligas Center*
 - Laguna Sports Complex (Sta. Cruz)
- **Rizal:**
 - Ynares Center (Antipolo City)
 - Montalban Sports Center (Rodriguez)
- **Quezon Province:**
 - Quezon Convention Center (Lucena City)

VII. REGION 4-B (MIMAROPA)

- **Palawan:**
 - Puerto Princesa City Coliseum
 - Palawan Convention Center
- **Oriental Mindoro:**
 - Calapan City Gymnasium
 - Provincial Capitol Complex (Calapan)
- **Marinduque:**
 - Marinduque Convention Center (Boac)

VIII. REGION 5 (BICOL REGION)

Priority: Typhoon resilience. High concentration of permanent DPWH centers.

- **Albay:**
 - Ibalong Centrum for Recreation (Legazpi City) – *Main Hub*
 - Albay Astrodome (Legazpi City)
 - Libon Evacuation Center (Sitio Tubog) – *Permanent DPWH Built*
 - Malinao Evacuation Center (Brgy. Balading) – *Permanent DPWH Built*
- **Camarines Sur:**
 - Capitol Convention Center (Pili)
 - Jesse M. Robredo Coliseum (Naga City)
 - Tinambac Evacuation Center (Brgy. Caliliog) – *Permanent DPWH Built*
- **Catanduanes:**
 - Virac Municipal Compound Evacuation Center – *Permanent DPWH Built*
 - Catanduanes State University Gym (Virac)
- **Sorsogon:**
 - Sorsogon Provincial Gymnasium (Sorsogon City)
 - Juban Evacuation Center (Brgy. Tughan) – *Permanent DPWH Built*
- **Masbate:**
 - Brgy. Bolod Evacuation Center (San Pascual) – *Permanent DPWH Built*

IX. GENERAL LOCAL LOGIC (TIER 2)

If the user is located in a Municipality not listed in Section II-VIII, the AI must infer the following probable sites:

1. **Primary Probability:** [Municipality Name] Municipal Gymnasium / Civic Center.
2. **Secondary Probability:** [Municipality Name] Central Elementary School.
3. **Tertiary Probability:** Barangay Covered Court / Hall.

Man made Disaster

MAN-MADE DISASTERS (Human-Induced Disasters)

Definition

A **man-made disaster** is a catastrophic event that occurs **as a direct or indirect result of human action**. These disasters can be caused by **carelessness, negligence, poor planning, industrial activity, conflict, or deliberate intent**. Unlike natural disasters, which are driven by nature, man-made disasters stem from **human mistakes or misuse of technology and resources**.

These events often lead to **loss of lives, destruction of property, environmental pollution, and long-term social and economic damage**.

TYPES OF MAN-MADE DISASTERS (with detailed descriptions)

1. VEHICULAR ACCIDENTS

Overview

A vehicular accident (or road traffic collision) happens when one or more vehicles collide with another vehicle, pedestrian, animal, or stationary object such as a tree, barrier, or building.

These accidents can cause injuries, deaths, and property damage, and are among the most common man-made disasters worldwide.

Vehicular accidents are typically preventable and result from human negligence, poor road design, or vehicle malfunction.

Main Causes

Human-Related Causes:

- **Overspeeding:** Driving beyond the speed limit reduces reaction time.
- **Distracted Driving:** Using phones, eating, or talking while driving.
- **Driving Under Influence (DUI):** Alcohol or drugs impair decision-making.
- **Drowsy Driving:** Sleep deprivation causes slow reactions and microsleep.
- **Reckless Behavior:** Tailgating, abrupt lane changes, ignoring signals.
- **Unlicensed or Untrained Drivers:** Lack of road knowledge and discipline.

Vehicle-Related Causes:

- **Mechanical Failures:** Brake or steering system failure.
- **Worn-Out Tires:** Reduced traction, causing skidding or blowouts.

- **Faulty Lights or Signals:** Increases collision risk at night or intersections.
- **Improper Vehicle Maintenance:** Ignoring scheduled servicing or inspections.

Environment & Road Causes:

- **Poor Road Design:** Sharp curves, narrow lanes, missing signs.
- **Bad Weather:** Rain, fog, or floods that obscure vision and traction.
- **Inadequate Lighting:** Dark or unmarked roads at night.
- **Obstacles on Roads:** Debris, potholes, or animals crossing unexpectedly.

Effects and Consequences

Human Impacts:

- Minor to fatal injuries (fractures, burns, internal bleeding, trauma).
- Loss of life among drivers, passengers, and pedestrians.
- Emotional and psychological distress (PTSD, depression, fear of travel).

Societal & Economic Impacts:

- Increased hospital and healthcare costs.
- Traffic congestion delaying emergency response and daily transport.
- Damage to infrastructure (lamp posts, sidewalks, barriers).
Loss of income and productivity for affected families.

Environmental Impacts:

- Fuel or oil spills contaminating soil and waterways.
- Vehicle fires releasing toxic gases into the air.

Classification of Vehicular Accidents

- **Head-On Collision** – Occurs when two vehicles crash front-to-front; often results in fatal injuries.
Rear-End Collision – Happens when one vehicle crashes into the back of another.
- **Side-Impact (“T-bone”) Collision** – One vehicle strikes the side of another, commonly at intersections.
- **Rollover Accident** – The vehicle flips onto its side or roof, usually caused by sharp turns or excessive speed.
- **Pedestrian Accident** – Involves a pedestrian being struck by a moving vehicle.
Multi-Vehicle Collision – A chain-reaction crash involving three or more vehicles.
- **Single-Vehicle Accident** – A vehicle collides with a stationary object such as a tree, pole, or barrier.

Before the Disaster (Preparedness & Prevention)

For Drivers:

- Perform **daily safety checks**: brakes, tires, headlights, wipers, engine fluids.
- Follow the **LTO driving code** and respect traffic signals.
- Wear seatbelts; ensure children use car seats.
- **Avoid alcohol or drugs** before driving.
- **Keep safe distance** between vehicles (“3-second rule”).
- Avoid using phones; use hands-free or pull over safely.
- Attend **defensive driving seminars** by accredited agencies.
- Be alert to **blind spots** and oncoming traffic.

For Pedestrians:

- Use **designated crosswalks and sidewalks**.
- Avoid crossing at blind corners or highways.
- Wear visible clothing at night.
- Refrain from phone use or wearing headphones while crossing roads.

For Government / LGUs:

- Regularly **inspect roads** for hazards and signage.
- Install **CCTV cameras** at accident-prone intersections.
- Conduct **public awareness campaigns** on road safety.
- Enforce strict penalties for reckless and drunk driving (RA 10586).
- Deploy **Traffic Enforcers and Barangay Tanods** for crowd control and assistance.

During the Disaster (Emergency Response)

If You Are Involved:

1. **Stay calm** — panic can worsen the situation.
2. Turn on **hazard lights** and pull over if possible.
3. **Check for injuries** — yourself, passengers, and others.
4. **Call emergency services** immediately:
 - **Emergency Hotline**: 911
 - **Police (PNP)**: 117
 - **Fire**: 160
 - **Rescue / DRRMO**: Local hotline
5. **Do not move injured victims** unless there’s immediate danger (e.g., fire, explosion risk).
6. **Use first aid** if trained.
7. Warn approaching drivers by placing **early warning devices** (reflectors) 5–10 meters behind your vehicle.
8. Cooperate with authorities upon arrival.

If You Witness an Accident:

- Park at a safe distance and activate hazard lights.
- Contact 911 or local police immediately.
- Report **exact location, number of vehicles, visible injuries, and road condition**.

- Avoid crowding or obstructing responders.
- Offer assistance only if safe and necessary.

After the Disaster (Recovery & Post-Response)

For Victims and Drivers:

- Get a **full medical examination**, even for minor injuries.
- Obtain a **police report** for legal and insurance purposes.
- Notify your **insurance provider** immediately.
- Seek **counseling or therapy** if experiencing trauma.
- Participate in community road safety education to help others.

For Authorities:

- Conduct **accident reconstruction and investigation** to identify causes.
- Submit data to **LTO / MMDA / DOTr** for national accident tracking.
- Review and improve **road safety policies** in high-risk areas.

Local and Global Data (as of latest records):

- The **World Health Organization (WHO)** reports **1.3 million deaths annually** due to road crashes.
- In the **Philippines**, according to **MMDA 2023 data**, there were over **58,447 recorded road accidents** in Metro Manila alone.
Motorcycles are involved in around **40% of all fatal road accidents** nationwide.
- **Peak accident hours:** 5:00 PM – 9:00 PM (rush hours).
- **Most accident-prone areas:** EDSA, Commonwealth Avenue, Marcos Highway, and SLEX.

Related Philippine Laws & Programs:

- **Republic Act No. 10586** – Anti-Drunk and Drugged Driving Act of 2013.
- **RA 4136** – Land Transportation and Traffic Code.
- **RA 11229** – Child Safety in Motor Vehicles Act.
- **DOTr Road Safety Program** – Advocates for zero road crash fatalities by 2030.

Educational Insights:

- Wearing seatbelts reduces the risk of death by **45–50%** in front-seat passengers.
- Helmets reduce fatal injuries by **42%** for motorcycle riders.
- **Texting while driving** increases crash risk by **23 times**.
- Following the **“3-second rule”** ensures proper distance to avoid rear-end collisions.

2. MINING AND QUARRYING DISASTERS

Overview

Mining and quarrying disasters occur during the extraction of minerals, rocks, or sand when unsafe conditions, equipment failure, or human error cause accidents. These disasters can result in **cave-ins, landslides, explosions, flooding, or exposure to toxic gases**, leading to severe injuries, fatalities, and environmental damage.

Causes

Human-Related Causes:

- Lack of proper **training for miners**.
- Ignoring **safety protocols** such as wearing protective gear.
- **Overworking** employees leading to fatigue and mistakes.
- **Illegal mining operations** without safety supervision.

Equipment & Technical Causes:

- Faulty machinery (drills, lifts, conveyors).
- Inadequate **ventilation systems**, leading to gas accumulation.
- Poorly maintained explosives used in blasting.
- Lack of monitoring for **structural stability** in tunnels.

Environmental & Site Causes:

- **Weak rock formations** causing collapses or landslides.
- Heavy rainfall or flooding leading to pit inundation.
- Earthquakes or tremors triggering tunnel collapse.
- Dust and particulate matter increasing respiratory hazards.

Effects and Consequences

Human Impacts:

- Fatalities due to **tunnel collapse, explosions, or drowning**.
- Severe injuries: fractures, burns, crush injuries, and respiratory problems.
- Long-term health issues: **pneumoconiosis (black lung), silicosis**, and hearing loss.

Environmental Impacts:

- Water contamination from chemicals or mining tailings.
- Soil degradation and erosion, leading to landslides.
- Habitat destruction for local wildlife.

Socio-Economic Impacts:

- Loss of jobs and income for miners and nearby communities.
- Expensive rehabilitation of mining sites.
- Local communities may lose access to water or farmland.
- Increased healthcare costs due to injuries and long-term diseases.

REAL-WORLD EXAMPLES

Tunnel Collapse / Cave-In – Roof or walls of tunnels fall due to poor support or weak rock.

Explosions – Detonation of explosives causing injuries and debris.

Flooding / Water Ingress – Mines or quarries flooded due to rain, dam breach, or groundwater.

Gas Leaks – Accumulation of methane, carbon monoxide, or other toxic gases.

Fires – Fires caused by sparks, electrical faults, or combustible material.

Equipment Accidents – Crushing, entrapment, or collisions with heavy machinery.

Before the Disaster (Preparedness & Prevention)

For Miners:

- Complete **safety training programs** including emergency evacuation.
- Always wear **PPE**: helmet, boots, gloves, dust mask, ear protection.
- Follow strict **blasting protocols** and chemical handling procedures.
- Maintain regular rest periods to avoid fatigue.
- Report any **cracks, water leaks, or unusual sounds** in tunnels.

For Mining Companies / Authorities:

- Conduct **regular structural inspections** of tunnels and pits.
- Install **ventilation systems** to remove toxic gases.
- Implement **emergency escape routes** and communication systems.
Monitor **weather conditions** and halt operations during heavy rains or storms.
- Conduct **mock drills** to ensure readiness for accidents.

During the Disaster (Response Phase)

If You Are Trapped or Injured:

- Stay calm and **conserve energy and air** if trapped.
- Try to **signal rescuers** using lights, sounds, or phones if available.
- Cover nose and mouth to reduce dust or smoke inhalation.
- Avoid moving heavy debris unless absolutely necessary.

If You Witness an Accident:

- Alert supervisors and **call emergency services immediately**.

- Keep clear of the danger zone (falling debris, flooding, fire).
- Assist injured miners **only if safe**.
- Do **not enter a collapsed tunnel** without proper rescue gear.

Emergency Hotlines (PH Example):

- **PNP / Police:** 117
- **Fire Department:** 160
- **Philippine Mines & Geosciences Bureau (MGB) Emergency:** Local office contact
- **DRRMO:** Local municipal hotlines

After the Disaster (Recovery Phase)

- Provide **medical treatment** and mental health support to survivors.
- Conduct **rescue and recovery operations** under professional supervision.
- Investigate the **cause of the disaster** and document findings.
- Repair and reinforce mining structures to prevent recurrence.
Ensure **safe relocation** for affected communities if environmental damage occurred.
- Educate workers and locals on lessons learned from the incident.

Educational Facts & Statistics

- Globally, mining accidents cause **thousands of deaths annually**, mostly in developing countries.
- The **2010 Chilean mine collapse** trapped 33 miners underground for **69 days**, all survived due to safety planning and international rescue support.
- In the Philippines, **illegal small-scale mining** contributes to many accidents, especially in remote mountainous areas.
- **Key hazards:** Dust inhalation (silicosis), toxic gases (methane, CO), and structural failure.

3. CYBER OR COMPUTER-RELATED DISASTERS

Overview

Cyber or computer-related disasters occur when critical computer systems, networks, or data are compromised, disrupted, or destroyed due to malicious attacks, technical failures, or human error. These disasters can affect individuals, businesses, and government institutions, causing financial loss, operational disruption, and reputational damage.

Cyber disasters are increasingly relevant in today's digital society, as more essential services depend on technology.

Causes

- **Malware and Viruses:** Malicious software that corrupts files, steals data, or disables systems.

- **Ransomware Attacks:** Hackers encrypt files or systems and demand payment for release.
- **Phishing and Social Engineering:** Users are tricked into revealing passwords or sensitive information.
- **Distributed Denial-of-Service (DDoS):** Flooding servers with traffic to make websites or services unavailable.
- **Insider Threats:** Employees or contractors intentionally or accidentally cause data breaches.
- **Hardware or Software Failure:** Servers, hard drives, or critical applications fail due to aging or misconfiguration.
- **Natural Disasters Affecting Data Centers:** Floods, fires, or earthquakes damaging hardware.
- **Weak Security Practices:** Poor password management, unpatched software, and lack of encryption.

Effects

- **Data Loss:** Permanent loss of personal, financial, or operational data.
- **Operational Disruption:** Companies, banks, or government offices may halt operations.
- **Financial Loss:** Costs include ransom payments, recovery, lost revenue, and regulatory fines.
- **Reputation Damage:** Loss of customer trust and negative media attention.
- **Legal Consequences:** Non-compliance with data protection laws can result in penalties.
- **Security Vulnerability:** Repeated attacks if system weaknesses are not addressed.
- **Psychological Impact:** Stress and anxiety among victims and employees.

Types of Cyber or Computer-Related Disasters

- **Ransomware Attack:** System or data is encrypted; hackers demand payment.
- **Data Breach:** Unauthorized access to sensitive information.
- **System Outage:** Failure of critical IT systems, servers, or networks.
- **Phishing Attack:** Fraudulent attempts to obtain user credentials or financial info.
- **Malware Infection:** Viruses, trojans, worms causing damage or theft.
- **Insider Threat:** Employees or contractors intentionally or accidentally compromise systems.
- **IoT Device Exploitation:** Internet-connected devices hacked to disrupt services.

Before the Disaster (Preparedness & Prevention)

- Regularly **backup data** to secure offline and cloud storage.
- Install and update **antivirus, anti-malware, and firewall software**.
- Conduct **employee cybersecurity training:** recognizing phishing, password security, and safe browsing.
- Use **strong, unique passwords** and enable multi-factor authentication (MFA).
- Regularly **update operating systems and software** to patch vulnerabilities.
- Develop a **cybersecurity incident response plan**, detailing communication and mitigation steps.

- Monitor systems for unusual activity using **network monitoring tools**.
- Ensure **physical security** of servers and data centers.

During the Disaster (Response Phase)

- **Immediately isolate affected systems** to prevent spread of malware.
- Contact **IT support or cybersecurity team** to assess damage.
- If ransomware, **do not pay the ransom** until professional advice is given; recovery might be possible from backups.
- **Notify affected users or clients** if sensitive data has been compromised.
- Activate the **incident response plan**.
- Record all **events and actions** for forensic investigation.
- If applicable, **report the incident** to authorities or data protection regulators (e.g., National Privacy Commission in the Philippines).

After the Disaster (Recovery & Post-Response)

- Restore data from **secure backups**.
- Conduct a **post-incident analysis** to identify vulnerabilities and prevent recurrence.
- Update **security protocols and software patches**.
- Provide **support to affected users** (credit monitoring, identity protection).
- Educate employees on **lessons learned** and reinforce cybersecurity training.
- Evaluate **insurance coverage** for cyber incidents.
- Implement **redundant systems and failover protocols** to minimize downtime in the future.

Educational Facts & Statistics

- Cybercrime cost businesses globally over **\$10 trillion** in 2023.
- The Philippines is considered one of the **most targeted countries in Southeast Asia** for phishing and online scams.
- Over **60% of data breaches** are caused by human error or weak passwords.
- Ransomware attacks can shut down critical services such as hospitals, banks, and government offices.
- Multi-factor authentication (MFA) can prevent up to **99.9% of account compromise attacks**.

3. FIRES AND EXPLOSIONS

Fires and explosions are man-made disasters that occur when **flammable materials, gases, or chemicals ignite** due to negligence, human error, equipment failure, or deliberate acts. These disasters can happen in **residential areas, commercial buildings, factories, fuel depots, or industrial plants**.

The impact can range from **minor property damage** to **mass casualties**, widespread destruction, and environmental hazards.

Causes

- **Human-Related Causes**
 - Negligence such as leaving stoves, heaters, or electrical appliances unattended.
 - Smoking in restricted areas or careless disposal of cigarette butts.
 - Mishandling flammable chemicals or fuels.
 - Arson or deliberate fire-setting.
- **Equipment-Related Causes**
 - Faulty electrical wiring, short circuits, or overloaded circuits.
 - Malfunctioning machinery in factories or industrial plants.
 - Improper storage of explosives, fuel, or chemicals.
- **Environmental & Structural Causes**
 - Lack of fire safety systems like sprinklers, smoke detectors, and alarms.
 - Poorly maintained buildings that accelerate fire spread.
 - Combustible construction materials increasing fire intensity.
- **Industrial & Gas Explosions**
 - Accidental ignition of **gas leaks** in pipelines, storage tanks, or refineries.
 - Improper handling or storage of explosives in mining, construction, or manufacturing.
 - Chemical reactions between incompatible materials.

Effects and Consequences

- **Human Impacts**
 - Injuries: burns, smoke inhalation, fractures, or crush injuries.
 - Fatalities due to intense heat, explosions, or trapped conditions.
 - Long-term health effects: respiratory diseases, burns requiring surgeries, psychological trauma.
- **Property and Economic Impacts**
 - Complete destruction of homes, factories, or infrastructure.
 - Loss of valuable equipment, inventory, or personal belongings.
 - Financial loss from business interruptions, repair costs, and insurance claims.

- **Environmental Impacts**
 - Air pollution due to smoke, toxic fumes, or chemical release.
 - Soil and water contamination from fire retardants, chemicals, or fuel spills.
 - Ecosystem damage if wildfires occur near forests or grasslands.

Types of Fires and Explosions

- **Residential Fires:** Kitchens, electrical appliances, or heating equipment.
- **Industrial Fires:** Factories, warehouses, or chemical plants.
- **Forest or Wildfires:** Dry vegetation ignited by human activity or sparks.
- **Gas Explosions:** Accidental ignition of LPG, natural gas pipelines, or fuel storage tanks.
- **Dust or Powder Explosions:** Combustible dust in factories, mills, or silos.
- **Chemical Explosions:** Uncontrolled reactions of chemicals or explosives.

Before the Disaster (Preparedness & Prevention)

- **For Individuals**
 - Install smoke detectors and fire alarms in homes and workplaces.
 - Keep fire extinguishers accessible and learn how to use them.
 - Develop an evacuation plan and practice fire drills regularly.
 - Avoid storing flammable materials near heat sources.
 - Ensure electrical wiring is inspected and maintained.
- **For Industrial Facilities**
 - Implement strict chemical handling and storage protocols.
 - Install automatic sprinkler systems, smoke detectors, and alarm systems.
 - Conduct regular **fire safety inspections**.
 - Train employees on **emergency response and firefighting techniques**.
 - Ensure clear and marked **emergency exits** and escape routes.
- **For Communities**
 - Maintain firebreaks in forested or grassland areas.
 - Conduct public awareness campaigns about fire hazards.
 - Ensure local fire departments are equipped and accessible.

During the Disaster (Response Phase)

- **If You Are In a Fire**
 - Stay low to avoid smoke inhalation; cover your nose and mouth with a damp cloth.
 - Evacuate immediately using **designated emergency exits**.
 - Do not use elevators in multi-story buildings.
 - Call emergency services (**Fire – 160 in the Philippines**) as soon as possible.
 - If trapped, signal for help from a window and stay near fresh air.
- **If You Witness an Explosion**
 - Move to a safe distance away from debris and secondary hazards.
 - Alert emergency responders and assist injured individuals only if safe.
 - Avoid approaching damaged structures that may collapse.
- **First Aid Measures**

- Treat minor burns with cool water and clean dressing.
- Stop bleeding from injuries using clean cloth or bandages.
- Avoid applying oils or creams to severe burns.
- Seek medical attention immediately for serious injuries.

After the Disaster (Recovery & Post-Response)

- **Medical and Psychological Support**
 - Treat burns, injuries, and smoke inhalation at hospitals.
 - Provide psychological support for trauma victims.
- **Property and Community Recovery**
 - Assess and repair damaged buildings with safety inspections.
 - Dispose of debris safely and prevent secondary fires.
 - Rebuild critical infrastructure and restore utilities.
- **Investigation and Prevention**
 - Conduct forensic investigation to determine the cause of fire or explosion.
 - Update safety protocols and employee training programs.
 - Review community or industrial safety regulations to prevent recurrence.

Educational Facts & Statistics

- Fires cause over **300,000 deaths worldwide** annually (WHO).
- In the Philippines, urban fires in residential areas and informal settlements are common due to **overcrowding, electrical faults, and open flame use**.
- Industrial explosions like the **Tianjin Port Explosion (2015)** highlight the dangers of unsafe chemical storage.
- Wildfires have destroyed **thousands of hectares** in forested regions, often accelerated by human activity.
- Proper fire safety training and smoke alarms can reduce **fatality risk by 50%** in residential fires.

4. Civil and Social Disasters

Overview

Civil and social disasters are **human-caused emergencies** that arise from conflict, mass panic, breakdown of public order, or sudden large gatherings of people. These disasters threaten community safety, disrupt social stability, and may result in injuries, deaths, or long-term trauma.

They include **riots, civil disturbances, terrorism, stampedes, mass protests turning violent, and large-scale public disorder events**.

Civil and social disasters are often unpredictable and influenced by cultural, political, or economic tensions.

Causes

- **Political Causes**
 - Electoral disputes and political unrest.
 - Government corruption or unpopular policies.
 - Power struggles between groups or factions.
- **Social/Economic Causes**
 - High unemployment and poverty.
 - Social injustice or discrimination.
 - Sudden price hikes of essential goods.
 - Public frustration over slow government service or crisis response.
- **Security-Related Causes**
 - Terrorist attacks or bomb threats.
 - Armed conflict between rebels and government forces.
 - Criminal activities causing mass panic.
- **Large Gatherings/Mass Events**
 - Religious gatherings, concerts, fiestas, or festivals.
 - Poor crowd control causing stampedes.
 - Overloaded venues without safety protocols.
- **Misinformation / Online Triggers**
 - Fake news spreading panic.
 - Social media posts inciting violence or chaos.
 - Cyber agitation leading to real-world unrest.

Effects and Consequences

- **Human Impacts**
 - Injuries from crowd crush, stampedes, gunfire, or physical assault.
 - Psychological trauma, fear, or long-term anxiety.
 - Loss of life during severe riots, terrorist attacks, or violent civil unrest.
- **Social Impacts**
 - Breakdown of community trust and social order.
 - Heightened tensions between groups (political, ethnic, or religious).
 - Displacement of families if violence escalates.
- **Economic Impacts**
 - Business closures, property damage, looting.
 - Loss of income for affected communities.
 - High cost of security and government response.
- **Infrastructure and Public Service Impacts**
 - Damage to roads, buildings, transportation, vehicles.
 - Shutdown of public services due to risk.
 - Increased burden on hospitals and emergency responders.

Types of Civil and Social Disasters

- **Riots** – violent disturbances with destruction of property.
- **Civil Unrest/Protests** – peaceful gatherings turning chaotic or violent.
- **Stampedes/Crowd Crush** – panic-driven movement causing mass injuries.
- **Terrorism** – bombs, armed attacks, chemical threats.
- **Armed Conflict** – clashes between military, rebels, gangs, or armed groups.
- **Mass Panic Events** – triggered by false alarms, misinformation, or sudden fear.
- **Public Disorder Situations** – looting, vandalism, mob attacks.

Before the Disaster (Preparedness & Prevention)

- **Government and Institutions**
 - Establish strong **public order and security protocols**.
 - Provide training for police on peaceful crowd management.
 - Monitor and fact-check misinformation sources.
 - Ensure proper emergency medical services during large events.
 - Plan traffic rerouting and crowd control barriers for gatherings.
- **Communities**
 - Stay informed through **verified news sources**.
 - Avoid participating in gatherings without proper security and permits.
 - Know the location of **safe exits** during big events.
 - Report suspicious individuals, activities, or unattended bags.
- **Individuals**
 - Prepare emergency contact numbers.
 - Stay away from high-risk areas during political tension.
 - Be aware of personal escape routes in crowded locations.
 - Practice situational awareness: observe surroundings and crowd behavior.

During the Disaster (Response Phase)

- **If caught in a riot or violent protest**
 - Move away from the center of conflict; avoid confrontation.
 - Do not run blindly; move diagonally out of the crowd.
 - Protect your head and vital organs.
 - Avoid wearing political or provocative clothing/colors.
- **If there is a stampede or crowd crush**
 - Keep your arms in front of your chest like a protective shield.
 - Move with the flow of the crowd, do not push against it.
 - If you fall, curl into a ball and protect your head until help arrives.
- **If terrorism or active shooter event**
 - Follow "Run, Hide, Fight" principles:
 - Run if safe.
 - Hide behind solid objects and lock doors.
 - Fight only as last resort using improvised weapons.
 - Stay silent; turn off phone sounds.
- **General Actions**

- Call emergency hotlines (PNP: 117 / local police).
- Avoid rumor spreading; follow official announcements.
- Assist injured individuals only if safe to do so.
- Follow evacuation routes and instructions from authorities.

After the Disaster (Recovery & Post-Response)

- **Medical and Psychological Support**
 - Get treatment for injuries such as fractures, cuts, or trauma.
 - Seek counseling if experiencing stress, anxiety, or panic attacks.
 - Hospitals may offer trauma intervention services after major events.
- **Community and Government Actions**
 - Cleanup operations to restore affected public areas.
 - Damage assessment and compensation for victims.
 - Strengthening social programs to address root causes.
 - Peace talks or dialogues between conflicting parties.
- **Legal and Investigative Actions**
 - Police and government conduct investigations.
 - Arrest or identify organizers of violent acts if applicable.
 - Document footage or evidence for legal cases.
- **Prevention for Future Events**
 - Improve crowd control systems for events.
 - Public awareness campaigns on peaceful protest participation.
 - Strengthen counter-terrorism intelligence and surveillance.
 - Online misinformation monitoring strategies.

Educational Facts & Insights

- Many civil disasters escalate due to **lack of crowd control** and poor communication.
- Crowd crush can occur in densities as low as **6–7 people per square meter**.
- Social media can increase unrest by spreading **unverified news** within minutes.
- Peaceful protests can turn violent due to **agitators, police response, or panic**.
- Terrorism incidents often target **public places** with high foot traffic.

5. ENVIRONMENTAL POLLUTION

Overview

Environmental pollution disasters occur when harmful substances contaminate **air, water, or land**, causing significant damage to human health, ecosystems, wildlife, and local economies. These disasters can be sudden (like oil spills) or gradual (like worsening air pollution or waste accumulation).

In the Philippines, environmental pollution impacts fisheries, tourism, agriculture, and urban communities.

Major Types of Environmental Pollution Disasters

1. Oil Spills

Oil spills occur when petroleum leaks into oceans, rivers, or coastal areas due to **ship accidents, pipeline leaks, or industrial negligence**.

2. Improper Waste Disposal

This includes the illegal dumping or mismanagement of solid waste, toxic chemicals, medical waste, and plastics.

3. Air Pollution

Air pollution happens when harmful gases, smoke, or particulate matter exceed safe levels—often due to **vehicle emissions, factory smoke, burning of trash, and industrial activities**.

Causes

- **Oil Spills**
 - Tanker accidents or ship collisions.
 - Pipeline ruptures or corrosion.
 - Illegal oil dumping or washing of ship tanks.
 - Offshore drilling accidents.
- **Improper Waste Disposal**
 - Lack of waste segregation and recycling.
 - Illegal dumping of garbage in rivers, streets, or open spaces.
 - Poorly managed landfills or open dumpsites.
 - Industrial and medical waste mismanagement.
- **Air Pollution**
 - Vehicle emissions (major contributor in Metro Manila).
 - Burning of plastics, leaves, or household waste.
 - Factories releasing toxic fumes or smoke.
 - Use of low-quality fuels in engines and generators.
 - Construction dust and urban congestion.

Effects and Consequences

- **Health Impacts**
 - Respiratory diseases (asthma, bronchitis, lung cancer).
 - Skin irritation and allergic reactions.

- Poisoning from contaminated water or fish.
- Increase in dengue, cholera, or diarrhea due to waste accumulation.
- **Environmental Impacts**
 - Death of marine life, birds, corals, and mangroves (especially in oil spills).
 - Contamination of soil and groundwater.
 - Loss of biodiversity.
 - Decline of air quality and smog formation.
- **Economic Impacts**
 - Fishermen losing income due to contaminated waters.
 - Tourism decline in polluted beaches and coastlines.
 - High cleanup and rehabilitation costs.
 - Medical expenses from pollution-related illnesses.
- **Community Impacts**
 - Evacuation or relocation due to toxic exposure.
 - Closure of schools and work interruptions.
 - Long-term damage to local livelihoods (e.g., fishing and farming).

Notable Example (Philippines)

- **Guimaras Oil Spill – 2006**
 - Caused by the sinking of MT Solar I.
 - Released **over 2 million liters of bunker fuel** into the sea.
 - Affected more than **1,000 hectares of mangroves** and **thousands of fishermen**.
 - Considered one of the **worst environmental disasters in Philippine history**.

Before the Disaster (Preparedness & Prevention)

- **Government and Industry**
 - Strict monitoring of tanker ships and ensuring they follow safety standards.
 - Regular inspection of pipelines and fuel depots.
 - Enforce waste segregation, recycling, and anti-littering laws.
 - Industries must install air filters and treat wastewater before disposal.
 - Develop emergency oil spill response teams and equipment.
- **Communities**
 - Participate in coastal cleanups and waste management programs.
 - Properly segregate household waste.
 - Report illegal dumping or smoke-belching vehicles.
 - Avoid burning plastics or garbage.
- **Individuals**
 - Reduce vehicle use by carpooling or using public transport.
 - Dispose of garbage properly and avoid single-use plastics.
 - Use reusable bags, bottles, and containers.
 - Plant trees and support local environmental initiatives.

During the Disaster (Response Phase)

- **Oil Spills**
 - Authorities deploy containment booms and skimmers.
 - Fisherfolk should avoid contaminated waters.
 - Evacuation may be required if fumes are toxic.
 - Monitor marine animals and prevent seafood consumption in affected zones.
- **Improper Waste Disposal Events**
 - Clear waterways and canals if flooding occurs due to blocked drainage.
 - Avoid direct contact with contaminated water or garbage piles.
 - Report large illegal dumping incidents to LGU or DENR.
- **Air Pollution**
 - Stay indoors if air quality becomes unhealthy.
 - Use masks when necessary.
 - Close windows and use air purifiers if available.
 - Avoid strenuous activities outdoors.

After the Disaster (Recovery & Post-Response)

- **Oil Spills**
 - Long-term marine rehabilitation (corals, fisheries, mangroves).
 - Government compensation for affected fishermen and communities.
 - Cleanup by trained responders and local volunteers.
 - Monitoring of marine ecosystems for months or years.
- **Improper Waste Disposal**
 - Community cleanup drives.
 - Implementation of stricter waste disposal ordinances.
 - Rehabilitation of polluted rivers, esteros, and canals.
- **Air Pollution**
 - Evaluate industrial sources and enforce stricter emission controls.
 - Implement policies like “no smoke-belching” and “carless days.”
 - Promote greener transportation and urban planning.

Educational Facts & Statistics

- Around **35% of air pollution in the Philippines** comes from vehicles (DENR).
- The Philippines produces **over 21 million tons of garbage per year**; only 30–40% is properly managed.
- Oil spills can destroy mangrove ecosystems, which may take **20–30 years** to fully restore.
- Burning plastic releases toxins like **dioxins**, which can cause cancer and organ damage.
- The Pasig River is one of the **most polluted rivers in the world** due to long-term improper disposal.

6. STRUCTURAL FAILURES

Definition

Structural failure refers to the **partial or complete collapse** of buildings, bridges, towers, roads, dams, or other infrastructures due to weakness, poor design, overloading, disasters, or lack of maintenance.

Common Causes

- **Poor Construction Materials**
 - Use of substandard cement, steel, or supports that cannot withstand load.
- **Design Flaws**
 - Incorrect structural calculations, weak foundations, or missing reinforcements.
- **Overloading**
 - Too many occupants, heavy equipment, or storage exceeding building capacity.
- **Lack of Maintenance**
 - Cracks, corrosion, and wear ignored for years weaken structural integrity.
- **Natural Forces**
 - Earthquakes, strong winds, flooding, soil erosion, liquefaction.
- **Illegal Construction or Modifications**
 - Unapproved renovations removing beams or walls that support the structure.
- **Soil Instability**
 - Building on reclaimed land, soft soil, or near slopes without proper evaluation.
- **Aging Infrastructure**
 - Old bridges, buildings, and roads without renovation programs.

Effects / Consequences

- **Loss of Lives & Injuries**
 - People trapped under debris, crush injuries, suffocation.
- **Displacement of Residents**
 - Evacuation needed for surrounding areas due to risk of further collapse.
- **Economic Losses**
 - Repair and rebuilding costs; business closures.
- **Traffic & Transportation Disruptions**
 - Bridge or road collapse affecting local or regional transport.
- **Damage to Utilities**
 - Broken water lines, power interruptions, or gas leaks.
- **Psychological Impact**
 - Trauma for survivors, especially children.
- **Legal & Government Action**
 - Demolition orders, investigations, suspensions of construction firms.

Examples in the Philippines

- **Ruby Tower Collapse (1968)**
Cause: Poor construction + intensity 7 earthquake
Result: 260 deaths, extensive reforms in building code.
- **Padre Burgos Bridge Collapse in Leyte (2022)**
Cause: Overloading by heavy trucks
Result: Bridge unusable, disrupted travel and trade.
- **Chuzon Supermarket Collapse in Porac, Pampanga (2019)**
Cause: Weak structural design; triggered by earthquake
Result: 11 deaths, full demolition ordered.

Risk Indicators You Should Watch For

- Visible cracks on walls, beams, or floors
- Sagging or uneven floors
- Doors and windows becoming hard to close
- Rusting steel frames
- Water leaks weakening concrete
- Unusual building vibrations
- Tilting of columns or posts

Before the Disaster (Preparedness Tips)

- Ensure your home/building is **inspected by a licensed structural engineer**, especially older buildings.
- Follow **National Building Code** and local construction standards.
- Avoid storing heavy items on upper floors unless the structure is rated for it.
- Do not remove load-bearing walls without professional approval.
- Report suspicious construction activities to local authorities.
- Prepare emergency kits:
 - Flashlight
 - Whistle (for alerting rescuers)
 - First-aid kit
 - Copies of important documents
- Know evacuation routes and safe open areas.

During the Disaster (If a Collapse Happens)

- If inside the building:
 - Protect your head with your arms or a solid object.
 - Stay away from windows, heavy shelves, and exterior walls.
 - Hide under a sturdy table (if available).
 - Do not use elevators.

- If trapped:
 - Stay calm and conserve energy.
 - Use a whistle or tap on pipes/walls to signal rescuers.
 - Avoid shouting continuously to save oxygen.
 - Cover nose and mouth to avoid inhaling dust.
- If outside:
 - Move away from buildings, power lines, and walls.
 - Warn others to avoid entering the structure.

After the Disaster (Recovery & Safety)

- Do not re-enter the collapsed or partially damaged structure.
- Wait for clearance from engineers or local authorities.
- Check for injured persons and provide first aid.
- Report gas leaks, water line breaks, and electrical sparks.
- Document damage for insurance or government assistance.
- Seek professional mental health support if needed.
- Participate in community cleanup only when declared safe.

How DisasterAlertBot Should Respond (Sample Output Format)

- Overview of structural failure
- Possible causes
- Warning signs
- Immediate recommended action
- Local emergency hotline
- Evacuation info (if available)
- Government resources (e.g., NDRRMC, DPWH reports)

Extra Useful Details for Chatbot Integration

- Add location-based alerts for:
 - Ongoing road or bridge closures
 - Faulty building inspection advisories
 - Earthquake-related structural damage
- Provide automatic tips for:
 - Renovation safety
 - Document storage before disasters
 - Engineering consultation reminders
- Provide school-friendly simplified explanations for students.

Typhoon And Tornado

What is a Typhoon?

- A typhoon is a powerful tropical cyclone that forms over the Northwest Pacific Ocean, specifically in the region west of the International Date Line and north of the equator. It is characterized by strong winds, heavy rainfall, storm surges, and flooding.

Key Facts

- Equivalent Terms:
 - Hurricane – in the Atlantic Ocean and Northeast Pacific.
 - Cyclone – in the Indian Ocean and South Pacific.
- Philippine Context:
 - The Philippines is hit by about 20 tropical cyclones each year, with around 8–9 making landfall.
 - The Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) monitors and issues warnings for typhoons.

Formation

- A typhoon cannot form just anywhere. Several specific conditions in the atmosphere and ocean must come together:

Requirement	Description	Why It's Important
Warm Ocean Water ($\geq 26.5^{\circ}\text{C}$)	The ocean's surface must be warm to a depth of about 50 meters.	Warm water provides the energy (heat and moisture) that fuels the storm.
Moist Air in the Mid-Troposphere	The middle layer of the atmosphere must have abundant water vapor.	Moisture condenses to form clouds and releases latent heat, which strengthens the system.
Low Vertical Wind Shear	Wind speed and direction must be relatively uniform from the ocean surface up to the upper atmosphere.	If wind shear is strong, it tears apart the forming cyclone.
Coriolis Force (Earth's Rotation)	The system must be far enough (about 5° latitude) from the equator.	Coriolis effect causes the system to rotate, forming a cyclonic spin.
Pre-existing Disturbance	Usually a cluster of thunderstorms or a low-pressure area.	Acts as the seed or starting point for the typhoon.

- Process

- Step 1: Warm Ocean Water Evaporates
 - The sun heats the ocean's surface.
 - Warm seawater evaporates and rises, carrying moist, warm air upward.
 - As the moist air rises, it cools and condenses into clouds, releasing latent heat energy.
 - *This heat makes the surrounding air warmer, allowing it to rise even more — starting a cycle of rising and condensing air.*
- Step 2: Formation of a Low-Pressure Area
 - As warm, moist air continues to rise, it leaves behind a low-pressure zone near the ocean surface
 - Air from surrounding areas (with higher pressure) moves in to replace it.
 - This incoming air warms, picks up moisture, and rises too — repeating the process.
 - This continuous cycle strengthens the low-pressure area and organizes it into a circular pattern.
- Step 3: Rotation Begins (Coriolis Effect)
 - The Coriolis force (caused by Earth's rotation) makes the rising air start to rotate:
 - Counterclockwise in the Northern Hemisphere (including the Philippines).
 - Clockwise in the Southern Hemisphere.
 - This rotation creates the early structure of a tropical depression.
- Step 4: Tropical Depression Stage
 - When wind speeds reach up to 61 km/h, the system becomes a Tropical Depression.
 - The structure is now more organized, with a circular shape and a developing core of low pressure.
- Step 5: Intensification into a Tropical Storm
 - As the system remains over warm waters, it continues to draw energy and moisture.
 - Wind speeds increase to 62–88 km/h, and it is now called a Tropical Storm.
 - The storm is given an official name by PAGASA (or the Japan Meteorological Agency, JMA).
- Step 6: Formation of the Eye and Eyewall
 - When winds reach 118 km/h or higher, it becomes a Typhoon.
 - The storm develops:
 - Eye – a calm, clear center where air sinks.
 - Eyewall – a ring of very strong thunderstorms around the eye; this is where the strongest winds and heaviest rains occur.
 - Spiral rainbands form outward from the center, bringing heavy rain and gusty winds far from the core.
 - The eye forms because intense upward motion in the eyewall causes air to rise rapidly, leaving a column of sinking, dry air in the center.

- Step 7: Maturity and Maximum Intensity
 - The typhoon reaches its peak strength when:
 - Sea surface remains warm.
 - Wind shear remains low.
 - The system remains over the ocean.
 - At this stage, it can reach “Super Typhoon” status with winds over 185 km/h.
- Step 8: Weakening and Dissipation
 - A typhoon weakens when it:
 - Moves over land (loses access to warm ocean water).
 - Encounters colder ocean waters.
 - Encounters strong vertical wind shear.
 - Reaches a dry air mass that cuts off moisture.
 - Once deprived of heat and moisture, the typhoon loses its energy and eventually dissipates.

Typhoon Formation Zones

- Most typhoons that affect the Philippines originate from:
 - The Pacific Ocean east of the Philippines (near the Caroline or Mariana Islands).
 - They travel westward or northwestward and may make landfall in the Philippines, then move toward Taiwan, China, or Japan.

Types of Typhoons (Based on Strength/Intensity)

- According to PAGASA and the World Meteorological Organization (WMO), tropical cyclones are classified based on maximum sustained wind speeds:

Classification	Wind Speed (km/h)	Description	Signal Number
Tropical Depression (TD)	≤ 61 km/h	Weak system with minimal damage potential; usually brings rain and wind.	1
Tropical Storm (TS)	62–88 km/h	Moderate system with stronger winds and heavier rains.	2
Severe Tropical Storm (STS)	89–117 km/h	Stronger winds and potential for moderate damage.	3
Typhoon (TY)	118–184 km/h	Powerful cyclone capable of severe damage to	4

		structures and vegetation.	
Super Typhoon (STY)	≥ 185 km/h	Extremely destructive; causes catastrophic damage, storm surges, and widespread flooding.	5

Rainfall Warning

- This advisory uses a color-coded system based on the total amount of expected rainfall:

Classification	Expected Rainfall (Hourly)	Description	Actions
Yellow (Be Alert)	7.5–15 mm	A Yellow Warning means rainfall is starting to intensify, and flooding is possible, especially in low-lying or flood-prone areas.	Stay alert and monitor official updates. Prepare important items in case evacuation becomes necessary. Check drainage, canals, and surroundings for possible blockages.
Orange (Be Prepared)	15–30 mm	An Orange Warning means flooding is threatening and may happen any moment as heavier rain continues.	Prepare to evacuate, especially if you live near rivers, creeks, or landslide-prone slopes. Secure your home, livestock, and valuables. Coordinate with barangay officials and be ready to move to safer ground.

Red (Take Action)	More than 30 mm	A Red Warning means serious and widespread flooding is expected or already occurring. This is the highest level of rainfall alert.	Evacuate immediately if instructed or if you feel unsafe. Move to designated evacuation centers or higher ground. Follow the guidance of local authorities and prioritize safety above belongings.
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What To Do Before, During, and After a Typhoon

A. Before a Typhoon

- Preparation is key to minimizing damage and ensuring safety.
 1. Stay Informed
 - Monitor weather bulletins via PAGASA, radio, TV, or official government pages.
 - Take note of Public Storm Warning Signals (PSWS) and their corresponding wind speeds and expected impacts.
 2. Prepare Your Home
 - Check and reinforce the roof, windows, and doors.
 - Trim trees and secure loose outdoor items.
 - Ensure drainage systems are not clogged.
 - Stock up on emergency supplies:
 - Food and water (good for 3 days)
 - Flashlight, extra batteries
 - First aid kit, medicines
 - Important documents (in waterproof containers)
 - Power banks or radios
 3. Plan for Evacuation
 - Know your nearest evacuation center.
 - Prepare an emergency “go bag” with essentials.
 - Secure pets and livestock.

B. During a Typhoon

- Stay indoors and prioritize safety.
 1. If at Home
 - Stay inside and keep away from windows or glass doors.
 - Turn off main electrical switches if flooding starts.
 - Avoid using electrical appliances.
 - Do not go outside even if the wind seems to stop — it could be the eye of the typhoon, and strong winds may return suddenly.
 2. If You Need to Evacuate
 - Follow local authorities' instructions.

- Move to higher ground if there is a threat of flooding or storm surge.
- Bring your emergency kit and valuables.
- Help the elderly, children, and people with disabilities.

C. After a Typhoon

- Stay alert even after the storm has passed.
 1. Safety Checks
 - Wait for official announcements declaring it safe to go outside.
 - Watch out for fallen power lines, debris, and floods.
 - Check your home for structural damage before re-entering.
 - Avoid wading in floodwaters (risk of electrocution or leptospirosis).
 2. Health and Sanitation
 - Boil drinking water if supply is contaminated.
 - Dispose of spoiled food properly.
 - Clean surroundings to prevent mosquito breeding.
 3. Stay Informed and Help Others
 - Listen for updates from authorities about relief and recovery efforts.
 - Assist neighbors, especially vulnerable groups.
 - Participate in community cleanup drives.

Devastating Typhoons in the History of the Philippines

Typhoon (Local / International Name)	Year	Approximate Peak Wind Speed	Estimated Damage / Losses
Yolanda / Haiyan	2013	315 km/h (1-min sustained, per JTWC)	₱ 95.5 billion
Rolly / Goni	2020	315 km/h	₱ 20 billion
Ompong / Mangkhut	2018	285 km/h	₱ 33.9 billion
Odette / Rai	2021	280 km/h	₱ 51.8 billion
Pablo / Bopha	2012	280 km/h	₱ 43.2 billion
Glenda / Rammasun	2014	260 km/h	₱ 38.6 billion
Pepeng / Parma	2009	250 km/h	₱ 27.3 billion
Ulysses / Vamco	2020	215 km/h	₱ 20.2 billion
Pedring / Nesat	2011	215 km/h	₱ 15.6 billion
Angela / Rosing	1995	285 km/h (1-min gust)	₱ 9.3 billion

What is a Tornado?

A tornado is a rapidly rotating column of air that extends from a thunderstorm to the ground. It is capable of causing severe destruction, including uprooting trees, destroying structures, and tossing vehicles. Tornadoes are less common in the Philippines than typhoons, but they do occur, often as part of severe thunderstorms or typhoon-induced weather.

Key Facts

- Equivalent Terms: Twister, cyclone (informally, but “tornado” is standard).
- Philippine Context: Tornadoes are rare but have been documented, mostly in the northern and central parts of Luzon and during the rainy season (June–November). PAGASA monitors severe thunderstorms that may produce tornadoes.
- Scale of Measurement: Fujita Scale (F-scale) or Enhanced Fujita Scale (EF-scale) estimates tornado strength based on damage.
- Are tornadoes common in the Philippines?
 - Yes, but usually weak.
 - The Philippines gets an average of 10–20 tornado events per year.
 - Most are EF0–EF1 (weak to moderate).
 - They occur most often during the rainy season (June–November) and are frequently associated with thunderstorms, typhoons, or localized wind convergence.
- Strongest / Notable Tornadoes in the Philippines
 - Iloilo Tornado (2023)
 - Affected Dumangas, Barotac Nuevo, and nearby towns
 - Damaged ~600 houses
 - Estimated at EF1–EF2 strength
 - Injured dozens
 - Manila / Quezon City Tornado (2016)
 - Hit Pulong Putik, Novaliches
 - Damaged multiple homes and structures
 - Estimated EF1
 - Laguna Tornado (2019)
 - Hit Biñan and Sta. Rosa
 - Threw metal roofing, toppled electric posts
 - Estimated EF1–EF2
 - Bulacan Tornado (2015)
 - Tore off roofs, downed trees
 - Around 300 families affected
 - Estimated EF1
 - Cavite Tornado (2012)
 - Damaged houses in Dasmariñas City
 - Estimated EF1
 - Cebu Tornado (2009)
 - Sheared roofs off buildings
 - Estimated EF1

Formation

A tornado requires several specific atmospheric conditions to form:

Requirement	Description	Why It's Important
Strong Thunderstorm / Supercell	A rotating thunderstorm (mesocyclone) is usually necessary.	Provides the energy and rotation needed to generate a tornado.
Warm, Moist Air	Warm, humid air near the surface fuels storm development.	Moisture feeds the thunderstorm, increasing instability.
Cold, Dry Air Aloft	Cold air above creates atmospheric instability.	Instability allows rising air to accelerate, forming strong updrafts.
Wind Shear	Changes in wind speed and direction with height.	Wind shear tilts the updraft, allowing rotation to develop and intensify.
Lift	A trigger, such as a cold front, typhoon outer bands, or terrain, to push air upward.	Helps thunderstorms to initiate and strengthens rotating updrafts.

Process of Tornado Formation

1. Thunderstorm Development
 - Warm, moist air rises rapidly, meeting cooler, drier air aloft.
 - Strong updrafts form a thunderstorm, often a supercell (rotating storm).
2. Mesocyclone Formation
 - Horizontal rotation due to wind shear is tilted vertically by the storm's updraft.
 - A rotating column of air forms inside the storm (mesocyclone).
3. Funnel Cloud Emergence
 - The rotation intensifies, forming a visible funnel cloud descending from the storm cloud base.
4. Tornado Touchdown
 - When the funnel cloud reaches the ground, it becomes a tornado.
 - The tornado may last from a few seconds to over an hour, depending on storm conditions.

Tornado Intensity (Fujita Scale / EF-Scale)

Classification	Wind Speed (km/h)	Description
F0 / EF0	105–137	Light damage, branches broken, minor roof damage.
F1 / EF1	138–178	Moderate damage, roofs peeled, mobile homes pushed.
F2 / EF2	179–218	Considerable damage, trees uprooted, large vehicles moved.
F3 / EF3	219–266	Severe damage, walls torn down, trains overturned.
F4 / EF4	267–322	Devastating damage, well-built houses leveled.
F5 / EF5	>322	Incredible damage, strong structures swept away.

Tornado Hazards in the Philippines

- Damage to houses, schools, and infrastructure.
- Uprooted trees and blocked roads.
- Flying debris causing injuries or fatalities.
- Localized flooding if combined with heavy rainfall.
- Often occurs in rural areas and can be triggered by typhoons or thunderstorms.

What to Do Before, During, and After a Tornado

Before a Tornado

1. Stay Informed
 - Monitor PAGASA Severe Weather Bulletins.
 - Be aware of thunderstorm watches and warnings.
2. Prepare Your Home
 - Secure loose outdoor items.
 - Identify a safe room (interior room without windows, basement if available).
3. Emergency Kit
 - Water, non-perishable food, flashlight, first-aid kit, batteries.
 - Helmet or heavy blankets to protect against debris.

During a Tornado

1. If Indoors
 - Go to the lowest floor and an interior room.
 - Avoid windows and doors.
 - Protect your head with a helmet, mattress, or heavy blankets.
2. If Outdoors
 - Seek shelter in a sturdy building.
 - If no building is available, lie flat in a low-lying area (ditch) and cover your head.
 - Do not seek shelter under bridges — wind speeds may increase.

After a Tornado

1. Safety Checks
 - Wait for official announcements before going outside.
 - Watch out for downed power lines, sharp debris, and unstable structures.
2. Health & Sanitation
 - Check for injuries and provide first aid.
 - Ensure drinking water is safe.
3. Community Support
 - Help neighbors, especially vulnerable populations.
 - Participate in clean-up and recovery efforts.

Tornado Occurrence Zones in the Philippines

- Northern and Central Luzon – Ilocos, Cagayan Valley, Central Luzon.
- Occasionally in Mindanao – Mostly during typhoon or severe thunderstorm events.
- Peak Season – June to November, coinciding with the rainy season and typhoon season.

Philippine Disaster Risk Management

- The Philippines' **Disaster Risk Reduction and Management (DRRM)** system is guided by:
 1. **Republic Act No. 10121 (Philippine DRRM Act of 2010)**
 - establishes the **National Disaster Risk Reduction and Management Council (NDRRMC)**.
 2. Focus areas:
 - **Disaster Prevention and Mitigation**
 - **Disaster Preparedness**
 - **Disaster Response**
 - **Disaster Rehabilitation and Recovery**
- Local governments and communities are trained to conduct:
 - Early warning dissemination
 - Evacuation drills
 - Post-disaster assessment and rehabilitation

Earthquake, Tsunami

EARTHQUAKE

What is an earthquake?

- An earthquake is a weak to violent shaking of the ground produced by the sudden movement of rock materials below the earth's surface.
- The earthquakes originate in the tectonic plate boundary. The focus is the point inside the earth where the earthquake started, sometimes called the hypocenter, and the point on the surface of the earth directly above the focus is called the epicenter.
- There are two ways by which we can measure the strength of an earthquake: magnitude and intensity. Magnitude is proportional to the energy released by an earthquake at the focus. It is calculated from earthquakes recorded by an instrument called seismograph. It is represented by Arabic Numbers (e.g. 4.8, 9.0). Intensity on the other hand, is the strength of an earthquake as perceived and felt by people in a certain locality. It is a numerical rating based on the relative effects to people, objects, environment, and structures in the surrounding. The intensity is generally higher near the epicenter. It is represented by Roman Numerals (e.g. II, IV, IX). In the Philippines, the intensity of an earthquake is determined using the PHIVOLCS Earthquake Intensity Scale (PEIS).

Types of Earthquake

- There are two types of earthquakes: tectonic and volcanic earthquakes. **Tectonic earthquakes** are produced by sudden movement along faults and plate boundaries. Earthquakes induced by rising lava or magma beneath active volcanoes is called **volcanic earthquakes**.

What causes earthquakes and where do they happen?

- The earth has four major layers: the inner core, outer core, mantle and crust. The crust and the top of the mantle make up a thin skin on the surface of our planet.
- But this skin is not all in one piece – it is made up of many pieces like a puzzle covering the surface of the earth. Not only that, but these puzzle pieces keep slowly moving around, sliding past one another and bumping into each other. We call these puzzle pieces tectonic plates, and the edges of the plates are called the plate boundaries. The plate boundaries are made up of many faults, and most of the earthquakes around the world occur on these faults. Since the edges of the plates are rough, they get stuck while the rest of the plate keeps moving. Finally, when the plate has moved far enough, the edges unstuck on one of the faults and there is an earthquake.

How are earthquakes recorded?

- Earthquakes are recorded by instruments called seismographs. The recording they make is called a seismogram. The seismograph has a base that sets firmly in the ground, and a heavy weight that hangs free. When an earthquake causes the ground to shake, the base of the seismograph shakes too, but the hanging weight does not. Instead the spring or string that it is hanging from absorbs all the movement. The difference in position between the shaking part of the seismograph and the motionless part is what is recorded.

Different Kind of Earthquake: Epicenter, Hypocenter, and Focus

1. *Focus (Earthquake Focus)*

- Definition

The focus is the exact point inside the Earth where the rocks first break or slip along a fault.

This is where the stored pressure or stress is suddenly released, producing seismic waves.

- Depth

The focus can be:

- Shallow (< 70 km deep) – most destructive
- Intermediate (70–300 km)
- Deep (300–700 km)

- Why Focus Matters

- Determines how strong the shaking will be on the surface.
- A shallow focus creates more intense ground shaking.
- A deep focus spreads energy over a wider area but causes weaker shaking.

- Example

The 2019 Mindanao earthquake had a shallow focus—this is why it caused strong ground shaking even though the magnitude wasn't extremely high.

2. *Hypocenter*

- Definition

The **hypocenter** is the **SAME** as the **focus**.

They both refer to the location *inside the Earth* where the earthquake starts.

- Why Two Words?

- **Focus** is used in general science.
- **Hypocenter** is often used in **seismology** and scientific reports.

- **Location**
 - The hypocenter is described using:
 - **Latitude**
 - **Longitude**
 - **Depth (km)**
- **How It Is Found**

Seismologists use **seismic waves recorded by seismographs** to calculate the hypocenter's depth and position.

3. Epicenter

- **Definition**
 - The **epicenter** is the point **on the Earth's surface** directly **above** the focus/hypocenter.
- **What It Represents**
 - The location reported in news and PHIVOLCS bulletins
 - The point usually associated with the strongest perceived shaking, though **not always** the most damaged location
- **How It Is Identified**

By **triangulation**:

1. Seismic stations measure the travel time of P-waves and S-waves.
2. The distance to the earthquake is calculated.
3. With data from 3 or more stations, they pinpoint the epicenter.

- **Example**

During the **2013 Bohol earthquake (M7.2)**:

- **Epicenter:** near Carmen, Bohol
- **Hypocenter:** several kilometers underground beneath the epicenter

Earthquake Scales: Types

A. MAGNITUDE SCALES

Magnitude scales measure the total energy released at the focus (hypocenter). They use numbers that do not depend on where you are—the magnitude is the same everywhere.

1. Richter Scale (Local Magnitude, ML)

- **Developed by:** Charles F. Richter (1935)
- **What It Measures:** The amplitude of seismic waves recorded by a seismograph.
- **Range:** Typically 0 to 9+, but it has limits—large earthquakes become inaccurate.
- **How It Works:** Uses a logarithmic system
- Each whole number increase = 10 times stronger shaking and 32 times more energy released
 - Example:**
 - A magnitude 6 vs. a magnitude 7:
 - 10 times more shaking
 - 32 times more energy
- **Limitations:**
 - Not accurate for large ($M > 7$) earthquakes
 - Only works well for California-type crust conditions

2. *Moment Magnitude Scale (M_w) – Modern Standard*

- **Developed by:** Kanamori, Hanks, and colleagues (1970s–1980s)
- **What It Measures:**
 - The seismic moment, which depends on:
 - Size of the fault area
 - Amount of slip (movement)
 - Rock strength
- **Why It's Best:**
 - Accurate for all earthquake sizes
 - Works globally
 - Used by PHIVOLCS, USGS, and all modern agencies
- **Range:** Usually 1.0 to 9.5+
- **Example:**
 - The 2011 Japan EQ is M_w 9.1
 - The 2019 Cotabato EQs were M_w 6.3–6.6
- **Fun Fact:** This scale replaced the Richter scale in scientific work.

3. Surface-Wave Magnitude (M_s)

- **Measures:** The amplitude of surface waves (Rayleigh waves)
Period: 20 seconds
- **Best Used For:** Medium to large shallow earthquakes
- **Issues:** Saturates at about M_s 8.0 (cannot measure bigger events properly)

4. Body-Wave Magnitude (M_b)

- Measures: The amplitude of body waves (P-waves) that travel deep underground
- Best For:
 - Deep earthquakes
 - Earthquakes > 300 km depth
- Limitations: Saturates for big earthquakes (cannot show true energy beyond M_w 6.5)

5. Energy Magnitude (M_e)

- **Measures:** Total radiated energy from an earthquake.
- **Purpose:** More accurate for comparing earthquakes that release different wave types.
- **Usage:** Mostly in scientific research.

B. INTENSITY SCALES

Intensity scales measure **how strong the shaking feels** and what **damage occurs**.

Intensity depends on:

- Distance from epicenter
- Soil type
- Building strength
- Earthquake depth

Unlike magnitude, intensity **changes by location**.

1. PHIVOLCS Earthquake Intensity Scale (PEIS) – *Used in the Philippines*

Range: Intensity I to X

Intensity Scale	Description
I	Scarcely Perceptible- Perceptible to people under favorable circumstances. Delicately balanced objects are disturbed slightly. Still Water in containers oscillates slowly
II	Slightly Felt – Felt by few individuals at rest indoors. Hanging objects swing slightly. Still, water in containers oscillates noticeably.
III	Weak – Felt by many people indoors especially in upper floors of buildings. Vibration feels like one passing of a light truck. Dizziness and nausea are experienced by some people. Hanging objects swing moderately. Still water in containers oscillates moderately.
IV	Moderately Strong – Felt generally by people indoors and by some people outdoors. Light sleepers are awakened. The vibration felt like a passing of heavy truck. Hanging objects swing considerably. Dinner, plates, glasses, windows and doors rattle. Floors and walls of wood framed buildings creak. Standing motor cars may rock slightly. Liquids in containers are slightly disturbed. Water in containers oscillate strongly. Rumbling sound may sometimes be heard.
V	Strong – Generally felt by most people indoors and outdoors. Many sleeping people are awakened. Some are frightened, some run outdoors. Strong shaking and rocking felt throughout the building. Hanging objects swing violently. Dining utensils clatter and clink; some are broken. Small, light and unstable objects may fall or overturn. Liquids spill from filled open containers. Standing vehicles rock noticeably. The shaking of leaves and twigs of trees are noticeable.

VI	Very Strong – Many people are frightened; many run outdoors. Some people lose their balance. Motorists feel like driving in flat tires. Heavy objects or furniture move or may be shifted. Small church bells may ring. Wall plaster may crack. Very old or poorly built houses and man-made structures are slightly damaged though well-built structures are not affected. Limited rockfalls and rolling boulders occur in hilly to mountainous areas and escarpments. Trees are noticeably shaken.
VII	Destructive – Most people are frightened and run outdoors. People find it difficult to stand on the upper floors. Heavy objects and furniture overturn or topple. Big church bells may ring. Old or poorly-built structures suffer considerable damage. Some well-built structures are slightly damaged. Some cracks may appear on dikes, fish ponds, road surface, or concrete hollow block walls. Limited liquefaction, lateral spreading and landslides are observed. Trees are shaken strongly. (Liquefaction is a process by which saturated sand lose strength during an earthquake and behave like liquid).
VIII	Very Destructive – People panicky. People find it difficult to stand even outdoors. Many well-built buildings are considerably damaged. Concrete dikes and foundation of bridges are destroyed by ground settling or toppling. Railway tracks are bent or broken. Tombstones may be displaced, twisted or overturned. Utility posts, towers and monuments may tilt or topple. Water and sewer pipes may be bent, twisted or broken. Liquefaction and lateral spreading cause man-made structure to sink, tilt or topple. Numerous landslides and rockfalls occur in mountainous and hilly areas. Boulders are thrown out from their positions particularly near the epicenter. Fissures and faults rupture may be observed. Trees are violently shaken. Water splash or stop over dikes or banks of rivers.
IX	Devastating – People are forcibly thrown to ground. Many cry and shake with fear. Most buildings are totally damaged. bridges and elevated concrete structures are toppled or destroyed. Numerous utility posts, towers and monument are tilted, toppled or broken. Water sewer pipes are bent, twisted or broken. Landslides and liquefaction with lateral spreadings and sandboils are widespread. the ground is distorted into undulations. Trees are shaken very violently with some toppled or broken. Boulders are commonly thrown out. River water splashes violently on slops over dikes and banks.

X	Completely Devastating – Practically all man-made structures are destroyed. Massive landslides and liquefaction, large scale subsidence and uplifting of land forms and many ground fissures are observed. Changes in river courses and destructive seiches in large lakes occur. Many trees are toppled, broken and uprooted.
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Why PEIS Is Important:

- Tailored to Philippine building types and soil conditions
- More accurate for local decisions and warnings

2. Modified Mercalli Intensity (MMI)

Range: I (not felt) to XII (total destruction)

Measures:

Effects on:

- People
- Buildings
- Natural features

Usage: Used globally before PEIS was developed; still used in many countries.

Example:

- MMI VII: “Difficult to stand; considerable damage in poorly built structures.”

3. Rossi–Forel Scale (Historical)

Range: I to X

Notes:

- Very old (late 1800s)
- No longer used today
- Replaced by MMI and PEIS

4. Japan Meteorological Agency (JMA) Scale

Range: 0 to 7

Features:

- Sharp, precise levels
- Commonly shown in Japan TV alerts
- Useful for rapid public warnings

FAULT LINES

Major Fault Systems & Their Locations

Philippine Fault Zone (PFZ)

- This is the longest and one of the most important fault systems in the country — roughly 1,200–1,250 km long, stretching from northwestern Luzon all the way down to southeastern Mindanao.
- It is a left-lateral strike-slip fault, which means the land on either side moves horizontally in opposite directions.
- It crosses many islands and is subdivided into many segments — Northern Luzon, Central Luzon, Visayas-Mindoro region, Mindanao — making it a major earthquake source across the archipelago.

Implication: Large parts of Luzon, Visayas, and Mindanao lie under the influence of this fault system. Earthquakes originating from PFZ can affect a wide geographic span.

Valley Fault System (VFS) - includes the West Valley Fault (WVF) and East Valley Fault (EVF)

- This is the major fault threat for Metro Manila and nearby provinces.
- West Valley Fault (WVF): Runs roughly north–south from Doña Remedios Trinidad, Bulacan → through Metro Manila → down to Laguna, Cavite, Batangas / Calamba. Its length is ~129–135 km.
- East Valley Fault (EVF): Shorter (about 17–30 km depending on source) — runs through parts of Rizal Province (e.g. Rodriguez, San Mateo) east of Metro Manila.
- VFS is considered capable of a major quake (up to ~M 7.0 -- 7.2), which could affect densely populated areas.

Implication: Metro Manila, Bulacan, Rizal, Laguna, Cavite, and nearby areas are at high risk if this fault moves — severe ground shaking, liquefaction, structural damage.

Lubang-Verde Passage Fault System (offshore between Mindoro and Batangas / Manila Trench area)

- This is an offshore strike-slip fault system that runs through the Verde Passage area (between Mindoro and the Batangas peninsula / Luzon).
- It accommodates part of the tectonic motion between plates in the region and remains an active seismic structure.

Implication: Coastal and offshore communities near Batangas, Mindoro — as well as nearby islands — may be at risk for earthquakes (and possible tsunami hazard if fault triggers undersea displacement).

Other Faults & Fault Lines (Regional / Lesser-known, but still Important)

There are many smaller or regional faults that — depending on location — may still pose seismic risk. A few examples:

- The Sibuyan Sea Fault — part of the broader PFZ — lies offshore north of Masbate, along the northeast edge of the Sibuyan Sea, passing near Marinduque and possibly affecting areas near Batangas peninsula.
- Various local fault systems listed in fault-line references include: West Ilocos Fault System, Abra River Fault, Casiguran Fault, Lubang Fault (branch of PFZ), Tablas Fault, Central Mindoro Fault, Mindoro-Lubang segments, and several others.
- In areas such as Bicol peninsula, there are fault splays (smaller fault branches) associated with PFZ's central segment (e.g. faults near volcanic zones) that have been identified by geologists.

Implication: Even if you're not near PFZ or VFS, local/regional faults may still produce potentially damaging earthquakes — so national coverage of fault lines matters across many provinces.

What This Means for Different Regions

Region / Area	Main Faults You Should Know About
Metro Manila + Bulacan, Rizal, Laguna, Cavite, Batangas	Valley Fault System (WVF & EVF) — highest urban risk.
Northern to Central Luzon, parts of Visayas, many islands including Mindanao	Philippine Fault Zone (PFZ) and its many segments — widespread risk across archipelago.
Mindoro–Batangas coast & Verde Passage area	Lubang-Verde Passage Fault System (offshore).
Coastal / offshore islands around Sibuyan Sea, Masbate, Marinduque, Batangas fringe	Sibuyan Sea Fault (branch of PFZ), other lesser-known offshore faults.
Various provinces (depending on location)	Regional / local faults (e.g. Abra River Fault, Casiguran Fault, etc.) — risk depends on proximity.

FACTS & TRIVIAS

Five Major Earthquakes in Philippine History:

1. 1990 Luzon Earthquake – Magnitude 7.7

Date: July 16, 1990

Affected Areas: Central and Northern Luzon (Baguio, Cabanatuan, Dagupan)

Key Facts:

- One of the strongest and most destructive earthquakes in modern PH history.
- Caused **liquefaction** in Dagupan and **massive landslides** in Baguio and Benguet.
- **Hyatt Terraces Hotel** collapsed.
- At least **1,621 people died**.
- Over **\$369 million** in damages.

Why It's Significant

This earthquake remains the standard reference for PH disaster planning due to its scale and impact.

2. 1976 Moro Gulf Earthquake – Magnitude 8.0

Date: August 17, 1976

Affected Areas: Moro Gulf, Mindanao

Key Facts:

- Triggered one of the **deadliest tsunamis** in PH history.
- Waves reached **4–9 meters high**.
- Struck communities in Cotabato, Sulu, Zamboanga, and Lanao.
- **Estimated 3,000–8,000 deaths**.

Why It's Significant

Its tsunami impact prompted improvements in Philippine coastal hazard monitoring.

3. 2013 Bohol Earthquake – Magnitude 7.2

Date: October 15, 2013

Affected Areas: Bohol, Cebu, Central Visayas

Key Facts:

- Destroyed **centuries-old Spanish-era churches**.
- Caused **landslides**, bridge collapses, and ground ruptures.
- **222+ fatalities** recorded.
- Over **73,000 structures damaged**.

Why It's Significant

It reshaped parts of Bohol's landscape and revealed an **active, previously unmapped fault** (the North Bohol Fault).

4. 1968 Casiguran Earthquake – Magnitude 7.3

Date: August 2, 1968

Affected Areas: Casiguran (epicenter), Manila heavily affected

Key Facts:

- Caused the collapse of **Ruby Tower** in Manila.
- More than **270 people died** in the Ruby Tower tragedy alone.
- Shaking felt across Central Luzon and Metro Manila.

Why It's Significant

This event influenced modern **Philippine building codes** and engineering safety standards.

5. 1645 Luzon Earthquake – Est. Magnitude 7.5

Date: November 30, 1645

Affected Areas: Manila, Pampanga, Nueva Ecija, Laguna

Key Facts:

- Known as “**Lindol na Dakila**” (Great Earthquake).
- Destroyed stone churches and buildings in Intramuros.
- Aftershocks lasted for **months**.
- Thousands died, although exact numbers are unclear.

Why It's Significant

It was the most destructive earthquake during the Spanish era and changed the architectural style of churches (Earthquake Baroque).

Interesting Trivia About Philippine Earthquakes

1. The Philippines has “Earthquake Swarm Zones.”

These are areas that shake repeatedly without major breaks:

- **Surigao City**
- **Bicol Region**
- **Mindoro**
- **Negros Oriental**

2. The strongest recorded earthquake in PH history was M8.6.

- The **1924 Mindanao earthquake** is considered one of the strongest seismic events in the country's recorded history.

3. The 1645 Luzon Earthquake was called the “Earthquake of the Spanish Era.”

- It destroyed churches, buildings, and even altered the course of rivers.

4. Earthquakes have reshaped PH landscapes.

Examples:

- **1968 Casiguran Earthquake** tilted buildings in Manila.
- **1990 Luzon Earthquake** created the famous “*Hanging Bridge Fault Scarp*” in Nueva Ecija.
- **Mindanao earthquakes** created ground ruptures several meters long.

5. Some Philippine lakes and rivers were formed because of earthquakes.

Example:

- **Lake Pinatubo** formed indirectly due to volcanic + tectonic activity.

Why the Philippines Has So Many Earthquakes

1. The Philippines Sits on the “Pacific Ring of Fire”

- This is a horseshoe-shaped belt around the Pacific Ocean where **90% of the world's earthquakes** occur.
- The country is affected by **multiple major tectonic plates**:
 - **Philippine Sea Plate** (to the east)
 - **Eurasian Plate** (to the west)
 - **Sunda Plate, Pacific Plate, Caroline Plate** (nearby)

2. Two Massive Fault Systems Cross the Country

Philippine Fault System (PFS)

- A **1,200 km long** system cutting from Luzon to Mindanao.
- Capable of producing **magnitude 7+ earthquakes**.
- Sections: Northern Sierra Madre, Bicol, Leyte, Surigao, Mindanao segments.

Valley Fault System (VFS)

- Located in Metro Manila and nearby provinces.
- Includes:
 - **West Valley Fault (WVF)** – the one capable of generating “**The Big One**” (M7.2).
 - **East Valley Fault** – smaller but active.

3. Daily Earthquakes

- The Philippines experiences **50–100 earthquakes every day**, most too weak to feel.
- Around **a few hundred noticeable earthquakes** happen annually.

EARTHQUAKE PREPAREDNESS

BEFORE AN EARTHQUAKE

Earthquakes strike without warning, so preparation is the key.

1. Prepare Your Home

- **Anchor tall furniture (cabinets, bookshelves) to the wall.**
- **Store heavy and breakable items on lower shelves.**
- **Place beds away from windows.**
- **Identify safe spots in every room:**

- Under sturdy tables or desks
- Beside interior load-bearing walls
- Away from glass, mirrors, and hanging objects

2. Prepare a Go-Bag (Emergency Kit)

Should be accessible near the door.

- Bottled water (at least 3 days)
- Ready-to-eat food (canned, biscuits, noodles)
- Flashlight + Extra batteries
- Whistle
- First-aid kit & personal medicines
- Radio (battery or hand-crank)
- Powerbank
- Important documents (in waterproof pouch)
- Clothes, blanket, hygiene kit
- Cash (ATMs may go offline)

3. Family Emergency Plan

- Establish 2 meeting points:
 - One near home
 - One outside your barangay
- Memorize emergency numbers:
 - NDRRMC: 143
 - Barangay Hotline
 - Local MDRRMO
- Teach kids how to do emergency calls even without load.
Assign a person to check elderly, children, and pets.

4. Strengthen Your House

- Have an engineer assess your home's structural integrity, especially older houses.

- Repair cracks in walls or foundations.
- Ensure gas tanks are secured and have no leaks.

5. Participate in Drills

Schools, offices, and malls conduct Earthquake Drills coordinated by:

- PHIVOLCS
 - LGU / Barangay
- Practicing “DROP–COVER–HOLD” makes your body react automatically during real quakes.

DURING AN EARTHQUAKE

Stay calm but act immediately.

If You’re Indoors

Perform DROP – COVER – HOLD:

- DROP to the ground
- COVER under a sturdy table or desk
- HOLD on until shaking stops

If no table:

- Protect head and neck with your arms
- Move next to an interior wall away from windows

Additional reminders:

- Stay away from windows, mirrors, shelves, appliances
- Do NOT run outside during shaking
- Avoid elevators

If You’re Outdoors

- Move to an open area away from:
 - Buildings
 - Trees
 - Power lines
 - Bridges

- **Drop low to the ground until shaking stops.**

If You're Driving

- **Pull over safely (not near trees, poles, bridges).**
- **Stay inside the vehicle until shaking stops.**
- **Drive carefully once roads are clear.**

If You're in a Crowded Place

- **Do not rush to exits.**
- **Protect your head and neck.**
- **Move away from falling objects (signboards, shelves).**

AFTER THE EARTHQUAKE

Expect aftershocks. Stay alert.

1. Check Yourself & Others

- **Treat injuries**
- **Call emergency services for severe cases**
- **Help elderly, children, and PWDs**

2. Inspect Your Home

Only do this when shaking stops.

Check for:

- **Gas leaks (if you smell LPG, evacuate immediately)**
- **Cracked walls, beams, floors**
- **Fallen electrical wires**
- **Damage to water pipes**

If unsafe → Evacuate and report to authorities.

3. Expect Service Interruptions

- **Power outages**
- **No phone signal**
- **Road blockages**

- **Water supply disruptions**

Listen to battery-powered radio or LGU announcements.

4. Stay Away From Dangerous Areas

- **Don't go near damaged buildings**
- **Avoid bridges until inspected**
- **Stay away from landslide-prone areas**

5. Prepare for Aftershocks

Aftershocks may continue for hours to months.

When they come, repeat DROP–COVER–HOLD.

6. If You Live Near the Coast

A strong or long earthquake (20+ seconds) may trigger a tsunami.

- **Immediately move to higher ground**
- **Do NOT wait for sirens or announcements**
- **Stay there until authorities declare "All Clear"**

TSUNAMI

What is a tsunami?

- A Tsunami is a series of sea waves commonly generated by under-the-sea earthquakes and whose heights could be greater than 5 meters. It is erroneously called tidal waves and sometimes mistakenly associated with storm surges. Tsunamis can occur when the earthquake is shallow-seated and strong enough to displace parts of the seabed and disturb the mass of water over it.
- There are two types of tsunami generation: Local *tsunami* and Far Field or distant tsunami. The coastal areas in the Philippines especially those facing the Pacific Ocean, South China Sea, Sulu Sea and Celebes Sea can be affected by tsunamis that may be generated by local earthquakes. Local tsunamis are confined to coasts within a hundred kilometers of the source usually earthquakes and a landslide or a pyroclastics flow. It can reach the shoreline within 2 to 5 minutes. Far field or distant tsunamis can travel from 1 to 24 hours before reaching the coast of the nearby countries. These tsunamis mainly

coming from the countries bordering Pacific Ocean like Chile, Alaska in USA and Japan. The Pacific Tsunami Warning Center (PTWC) and Northwest Pacific Tsunami Advisory Center (NWPTAC) are the responsible agencies that closely monitor Pacific-wide tsunami event and send tsunami warning to the countries around the Pacific Ocean.

- The Philippines is frequently visited by tsunamis. On 17 August 1976, a magnitude 8.1 earthquake in Moro Gulf produced up to 9-meter high tsunamis which devastated the southwest coast of Mindanao and left more than 3,000 people dead, with at least 1,000 people missing. Also on 15 November 1994 Mindoro Earthquake also generated tsunamis that left 49 casualties.
-

BEFORE A TSUNAMI

1. . Know Your Risk

- Check if your home, school, or workplace is in a **tsunami-prone area**.
- Visit or ask your LGU for **hazard maps**.
- Live near beaches, coastal barangays, or ports? You are at higher risk.

2. . Learn Evacuation Routes

- Identify multiple routes going to **higher ground** (at least 30 meters above sea level).
- Know the **nearest evacuation center**.
- Practice walking the route with your family, especially kids and seniors.

3. Prepare an Emergency Go-Bag

A grab-and-go bag must include:

- Water (3-day supply)
- Ready-to-eat food
- Flashlight + batteries
- Whistle
- First-aid kit
- Powerbank
- Radio (battery or hand-crank)
- Important documents in waterproof pouch
- Extra clothes and hygiene items
- Cash

4. Understand Warning Signs

A tsunami may follow:

- A **strong or long-lasting earthquake**
- The **sea rapidly receding**
- A **loud, roaring ocean sound**

- Official warnings from:
 - PHIVOLCS
 - NDRRMC
 - Coast Guard
 - LGUs

5. Make a Family Disaster Plan

- Establish a **meeting point** after evacuation.
- Teach children what to do if an earthquake happens at school or home.
- Save emergency hotlines (NDRRMC: 143).

DURING A TSUNAMI (RESPONSE)

1. Right After a Strong or Long Earthquake

A strong quake is a **natural tsunami warning**.

Evacuate immediately to high ground — DO NOT wait for announcements.

2. If You See the Sea Recede Suddenly

This is one of the clearest signs of an incoming tsunami.

Run inland or uphill immediately.

3. During an Official Tsunami Warning

- Move quickly to the evacuation site.
- If roads are crowded, go **on foot** to avoid being trapped in traffic.
- Do NOT return to the shore to take photos or videos.
- If you are at sea, **stay in deep water**—returning may be more dangerous.

4. If You Are Indoors

- Evacuate immediately
- Bring your go-bag
- Do not waste time gathering belongings

5. If You Cannot Evacuate in Time

Move to the **third floor or higher** of a sturdy building.

Avoid basements.

AFTER A TSUNAMI (RECOVERY)

1. Stay in High Ground

- Wait for the official “**All Clear**” from PHIVOLCS or LGU.
- Remember: **multiple waves** can arrive hours apart.

2. Avoid Dangerous Areas

Do NOT return to:

- Flooded zones
- Damaged buildings
- Broken bridges
- Areas with fallen wires
- Contaminated water

3. Check for Injuries

- Give first aid if trained.
- Assist children, elderly, and those with disabilities.

4. Report Missing Persons

- Go to the barangay command center or LGU desk.
- Check official lists of rescued or missing individuals.

5. Protect Your Health

- Do not drink tap water until declared safe.
- Avoid stepping into floodwater to prevent infection or electrocution.

6. Document Damages

- Take photos of house or property damage for insurance or LGU assistance.
- Register with DSWD or barangay for relief and support.

Volcanic Eruption, Wildfires

Volcanic Eruptions

A volcanic eruption occurs when magma, gas, and ash escape from beneath the Earth's crust through vents, fissures, or craters. Eruptions can be explosive or effusive and may release lava flows, ashfall, pyroclastic density currents, volcanic bombs, and toxic gases.

Volcanic eruptions are among the most destructive natural disasters, capable of affecting entire regions through ashfall, lahars, earthquakes, and long-term environmental change.

The Philippines is part of the Pacific Ring of Fire, hosting 24 active volcanoes, including Taal, Mayon, and Pinatubo.

Causes

Geological Causes

- Magma Pressure Build-up: Accumulation of molten rock beneath the crust.
- Movement of Tectonic Plates: Subduction zones melt rock and generate magma.
- Gas Expansion: Dissolved gases in magma increase pressure until it erupts.
- Cracks & Weak Points: Fractures allow magma to rise rapidly.

Triggering Factors

- Small-to-moderate earthquakes.
- Hydrothermal explosions due to water contacting hot magma.
- Reactivation from previous eruptions or magma intrusions.
- Landslides exposing hot material.

Effects and Consequences

Human Impacts

- Fatalities from pyroclastic flows, ashfall roof collapse, lahars, and toxic gases.
- Respiratory issues due to ash inhalation (asthma attacks, lung inflammation).
- Burns, eye irritation, and injuries from falling debris.

Environmental Impacts

- Ashfall damaging crops, forests, coral reefs.
- River blockages causing long-term flooding risks.
- Landscape changes, crater formation, and altered water systems.

Socio-Economic Impacts

- Loss of homes and infrastructure.
- Flight cancellations and airport closures.
- Agricultural and livestock destruction.
- Long-term displacement of communities.

Types of Volcanic Hazards

- **Lava Flows** – slow but destructive to structures.
- **Ashfall** – spreads hundreds of kilometers, contaminates water, damages lungs.
- **Pyroclastic Density Currents** – fast-moving, extremely hot clouds of ash and gas.
- **Lahars (Volcanic Mudflows)** – mixtures of ash, debris, and rainwater that flow rapidly down rivers.
- **Volcanic Gas** – sulfur dioxide, carbon dioxide, and hydrogen sulfide.
- **Ground Deformation** – swelling or sinking of land.
- **Volcanic Earthquakes** – shaking due to magma movement.

Before the Disaster (Preparedness)

For Government / LGUs:

- Monitor volcanoes via PHIVOLCS.
- Establish exclusion zones based on Alert Levels.
- Prepare evacuation centers away from lahar paths.
- Conduct regular community drills.
- Issue real-time advisories on ashfall and volcanic tremors.

For Communities & Individuals:

- Know PHIVOLCS Alert Levels and meanings.
- Prepare emergency kits:
 - Masks (N95 recommended), goggles
 - Food & water (3 days)
 - Flashlight, batteries
- Strengthen roofs against ash accumulation.
- Keep animals and livestock in safe areas.
- Identify evacuation routes and nearby shelters.

During the Disaster (Response)

If there is ashfall:

- Stay indoors; close windows and doors.
- Wear masks and protective goggles.
- Avoid driving—ash reduces visibility and damages engines.
- Cover water containers and food.

If Near the Volcano

- Evacuate immediately when instructed.
- Keep away from rivers that may carry lahars.
- Avoid low-lying areas susceptible to pyroclastic flows.
- Bring emergency kits and important documents.

Emergency Hotlines (PH)

- PHIVOLCS: 02 8 426 1468 to 79
- NDRRMC: 911
- Fire: 160
- PNP: 117

After the Disaster (Recovery)

- Remove ash using damp cloths; avoid sweeping dry ash.
- Check structures for roof damage.
- Ensure drinking water is safe.
- Dispose of ash properly and do not dump it in waterways.
- Attend medical checkups for respiratory issues.
- Support community clean-up efforts.

Notable PH Examples

- **Mount Pinatubo (1991)** – second largest eruption of the 20th century.
- **Taal Volcano (2020)** – widespread ashfall, mass evacuations.
- **Mayon Volcano (multiple years)** – ongoing lava flows and pyroclastic activity.

Educational Insights

- Fine ash can travel hundreds of kilometers from the eruption site.
- Wet ash can be heavier than cement, collapsing roofs.
- Pyroclastic flows can reach speeds of 100–700 km/h and temperatures of 1,000°C.
- Volcanic soil becomes extremely fertile long-term.

WILDFIRES (Forest and Grassland Fires)

A wildfire is an uncontrolled fire that spreads across forests, grasslands, agricultural fields, or rural communities. While some wildfires begin naturally (e.g., lightning), most are human-caused, such as unattended fires, burning of trash, or arson.

Wildfires can destroy ecosystems, homes, crops, and infrastructure, and produce smoke that affects air quality over large regions.

In the Philippines, wildfires commonly occur in mountainous regions, grasslands, and dry-season forests.

Causes

Human-Related Causes

- Kaingin (slash-and-burn farming).
- Unattended campfires or bonfires.
- Burning garbage or agricultural waste.
- Cigarette butts thrown on dry grass.
- Sparks from machinery or vehicles.
- Arson.

Natural Causes

- Lightning strikes.
- Extremely hot and dry weather.
- Prolonged drought associated with El Niño.
- Environmental & Structural Causes
- Accumulation of dry leaves, brush, or dead vegetation.
- High winds spreading embers.
- Steep slopes accelerating fire movement.

Effects and Consequences

Human Impacts

- Burns, smoke inhalation, respiratory distress.
- Evacuations and loss of homes.
- Increased risk of asthma or chronic lung problems.

Environmental Impacts

- Destruction of wildlife habitats.
- Soil erosion and loss of forest cover.
- Increased risk of landslides and flooding after rainfall.
- Air pollution and reduced visibility.

Socio-Economic Impacts

- Damage to farmland and livestock.
- Closure of parks, tourism sites, and mountain trails.
- Costly firefighting efforts and reforestation programs.

Types of Wildfires

- **Surface Fires:** Burn leaves, grass, shrubs.
- **Crown Fires:** Ignite the tops of trees—extremely fast and destructive.
- **Ground Fires:** Burn organic soil layers (peat), long-lasting and hard to extinguish.

Before the Disaster (Preparedness & Prevention)

For Individuals & Communities

- Avoid burning trash during dry season.
- Clear dry leaves and flammable debris from surroundings.
- Create defensible space around homes (no dry brush).
- Prepare emergency kits and masks.
- Report smoke or illegal burning to authorities immediately.

Government / LGUs

- Monitor fire-prone zones during hot/dry months.
- Enforce anti-burning laws (RA 9003 / RA 8749).
- Conduct firebreak creation and reforestation.
- Set up community fire brigades and volunteer groups.
- Install warning signages in protected areas.

During the Disaster (Response Phase)

If You Are Near a Wildfire

- Evacuate immediately following LGU instructions.
- Move downhill—fires spread upward faster than downward.
- Cover your nose and mouth with a wet cloth or mask.
- Keep low to avoid smoke inhalation.

If Trapped

- Find an area with minimal vegetation.
- Lie face-down and cover your body with soil or non-flammable material.
- Call emergency services.

Hotlines

- Fire Department: 160
- LGU DRRMO
- DENR for forest areas

After the Disaster (Recovery)

- Wait for official clearance before returning home.
- Watch out for falling branches or weakened trees.
- Check structures for fire damage.
- Purify water if supply is contaminated by ash.
- Participate in reforestation and soil stabilization efforts.
- Monitor air quality for several days after the fire.

Philippine Examples

- **Sierra Madre wildfires** (dry season, recurring annually).
- **Mt. Apo wildfire (2016)** – damaged 300+ hectares.
- **Baguio / Benguet** grassfires during El Niño years.

Educational Insights

- 90% of wildfires worldwide are caused by human activity.
- Fires spread faster uphill due to rising heat.
- Fine wildfire smoke (PM2.5) can travel hundreds of kilometers.
- Reforestation after wildfire is critical to prevent landslides.

Disaster Preparedness & First Aid Handbook

Prepared by the Senate Committee on Climate Change in coordination with the Philippine Institute of Volcanology and Seismology (PHIVOLCS), National Disaster Risk Reduction and Management Council (NDRRMC), Department of Education (DepEd), Department of Health (DOH), Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA), Bureau of Fire Protection (BFP), Mines and Geosciences Bureau (MGB) and Vigan City Disaster Risk Reduction and Management Council.

A project of the Senate Committee on Climate Change in cooperation with the Department of Environment and Natural Resources

EARTHQUAKE

WHAT IS AN EARTHQUAKE?

An earthquake is a feeble shaking to violent trembling of the ground produced by the sudden displacement of rocks or rock materials below the earth's surface. Tectonic earthquakes are generated by the sudden displacement along faults in the solid and rigid layer of the earth. Volcanic earthquakes are induced by rising magma beneath active volcanoes. THE PHILIPPINES AS AN EARTHQUAKE- PRONE COUNTRY

The Philippine Archipelago lies between 2 major tectonic plates, the Philippine Sea Plate and the Eurasian Plate. The Philippine Sea Plate is being subducted westward along the Philippine Trench at the eastern side of the Philippines. The Eurasian Plate is being subducted eastward along the Manila Trench, Negros-Sulu Trench and the Cotabato Trench at the

western side of the country except in southern Mindoro and western Panay, where collision takes place. Active faults traverse the archipelago, the longest of which is the Philippine Fault Zone. Subduction of plates along the trenches and movements along active faults are responsible for the present-day high seismicity of the Philippines.

For the last 400 years, the Philippines has been affected by 90 damaging earthquakes. Hence, the likelihood of destructive earthquakes occurring again in the future is indeed very strong. Around 20 earthquakes per day occur in the Philippines, most of which are not felt. About two hundred earthquakes are felt every year.

HAZARDS POSED BY EARTHQUAKES

1. GROUND SHAKING

The destructive effects of earthquakes are due mainly to intense GROUND SHAKING or vibration. Because of severe ground shaking, low and tall buildings, towers, and posts may tilt, split, topple or collapse; road foundations, railroad tracks and bridges may break; water pipes and other utility installations may get dislocated; dams and similar structures may break and cause flooding; and other forms of mass movement may be generated.

2. LIQUEFACTION

LIQUEFACTION is a process where particles of loosely consolidated and water-saturated deposits of fine sand are

rearranged into a more compact state. Water and sediments are squeezed out towards the surface in the form of water and sand fountaining (sand boiling), thus creating a condition resembling 'quick sand'. The consequent loss in volume and underlying support results in subsidence of the ground on top of the liquefying sandy layers and, with it, the sinking and/or tilting of any

structures above it. Liquefaction prone areas may be found in beach zones, sand spits, sand bars, tombolos, wide coastal plains, deltaic plains, floodplains, abandoned river meanders, former lake beds, former or existing marshlands and swamplands, and in areas underlain by sandy lahar deposits.

3. LANDSLIDES

Landslides are a downward movement of slope materials either slowly or quickly. A landslide may be a rock fall, topple, slump, slide or debris flow. Intense ground shaking can trigger a landslide by loosening the cohesion that bonds the slope materials together, thereby making it easier for gravity to pull it downwards. Hilly and mountainous areas, escarpments, steep river banks, sea cliffs, and other steep slopes are prone to landsliding. The main effect of landsliding is burial.

4. GROUND RUPTURE

Many strong earthquakes originate along faults that break the earth's rigid crust. GROUND RUPTURE is a deformation on the ground that marks the intersection of the fault plane with the earth's surface. The most common manifestation is a long fissure extending from a few kilometers to tens of kilometers, although ground rupture may also occur as a series of discontinuous cracks, mounds, or depressions.

The length of ground rupture and the width of the zone of deformation generally increase with the magnitude and type of earthquake.

5. TSUNAMI

Tsunamis are small to giant sea waves mostly generated by submarine earthquakes. Not all submarine earthquakes, however, can cause tsunamis to occur. Tsunamis can only occur when the earthquake is shallow-seated, and strong enough (about Magnitude 7) to displace parts of the seabed and disturb the mass of water over it. Other causes of tsunamis include submarine or coastal landslides, pyroclastic flows, large volume debris avalanches from submarine and partly submerged volcanoes, and caldera collapse.

PHIVOLCS EARTHQUAKE INTENSITY SCALE

Intensity I: Scarcely Perceptible - Perceptible to people under favorable circumstances.

Delicately balanced objects are disturbed slightly. Still water in containers oscillates slowly.

Intensity II: Slightly Felt - Felt by few individuals at rest indoors. Hanging objects swing slightly.

Still water in containers oscillates noticeably.

Intensity III: Weak - Felt by many people indoors especially in upper floors of buildings. Vibration is felt like one passing of a light truck. Dizziness and nausea are experienced by some people. Hanging objects swing moderately. Still water in containers oscillates moderately.

Intensity IV : Moderately Strong - Felt generally by people indoors and by some people outdoors. Light sleepers are awakened.

Vibration is felt like the passing of a heavy truck. Hanging objects swing considerably. Dinner plates, glasses, windows, and doors rattle. Floors and walls of wood-framed buildings creak. Standing motor cars may rock slightly. Liquids in containers are slightly disturbed. Water in containers oscillates strongly.

Rumbling sound may sometimes be heard.

Intensity V: Strong -Generally felt by most people indoors and outdoors. Many sleeping people are awakened. Some are frightened, some run outdoors. Strong shaking and rocking felt throughout building. Hanging objects swing violently. Dining utensils clatter and clink; some are broken. Small, light, and unstable objects may fall or overturn. Liquids spill from filled open containers. Standing vehicles rock noticeably. Shaking of leaves and twigs of trees are noticeable.

Intensity VI:

Very Strong

- Many people are frightened; many run outdoors. Some people lose their balance. Motorists experience a sensation of driving on flat tires. Heavy objects or furniture move or may be shifted. Small church bells may ring. Wall plaster may crack. Very old or poorly built houses and man-made structures are slightly damaged, though well- built structures are not affected. Limited rockfalls and rolling boulders occur in hilly to mountainous areas and escarpments. Trees are noticeably shaken.

Intensity VII: Destructive - Most people are frightened and run outdoors. People find it difficult to stand in upper floors. Heavy objects and furniture overturn or topple. Big church bells may ring. Old or poorly- built structures suffer considerable damage. Some well-built structures are slightly damaged. Some cracks may appear on dikes, fish ponds, road surface, or concrete hollow block walls. Limited liquefaction, lateral spreading and landslides are observed. Trees are shaken strongly.

Intensity VIII: Very Destructive

- People are panicky. People find it difficult to stand, even outdoors. Many well-built buildings are considerably damaged. Concrete dikes and foundation of bridges are destroyed by ground settling or toppling. Railway tracks are bent or broken. Tombstones may be displaced, twisted or overturned. Utility posts, towers and monuments may tilt or topple. Water and sewer pipes may be bent, twisted or broken. Liquefaction and lateral spreading cause man-made structures to sink, tilt, or topple. Numerous landslides and rockfalls occur in mountainous and hilly areas. Boulders are thrown out from their positions, particularly near the epicenter.

Fissure- and fault-rupture may be observed. Trees are violently shaken. Water splashes or slops over dikes or banks of rivers.

Intensity IX: Devastating

- People are forcibly thrown to the ground. Many cry and shake with fear. Most buildings are totally damaged. Bridges and elevated concrete structures are toppled or destroyed. Numerous utility posts, towers, and monuments are tilted, toppled, or broken. Water sewer pipes are bent, twisted, or broken. Landslides and liquefaction with lateral spreadings and sandboils are widespread. The ground is distorted and undulates. Trees are shaken very violently, with some toppled or broken. Boulders are commonly thrown out. River water splashes violently or slops over dikes and banks.

Intensity X: Completely Devastating - Practically all man-made structures are destroyed. Massive landslides and liquefaction, large scale subsidence and uplifting of land forms, and many ground fissures are observed. Changes in river courses and destructive seiches in large lakes occur. Many trees are toppled, broken and uprooted.

WHAT TO DO BEFORE, DURING, AND AFTER AN EARTHQUAKE

Before

1. The key to effective disaster prevention is planning.

Determine whether the site is along an active fault and/or prone to liquefaction or landslide which may cause damage to your house or building.

Be sure that proper structural design and engineering practice is followed when constructing a house or building.

Evaluate the structural soundness of buildings and important infrastructure; strengthen or retrofit if found necessary.

Plan ahead with your family in case of an earthquake. Decide on a place to meet in the event that you are separated.

Develop family, workplace and community preparedness and evacuation plans

2. Prepare your place of work and residence for the event.

Strap heavy furniture/cabinets to the wall to prevent sliding or toppling.

Check the stability and secure hanging objects, like ceiling fans and chandeliers, which may break loose and fall during earthquakes.

Heavy materials, breakable items, harmful chemicals, and flammable materials should be stored in the lowermost shelves and secured firmly.

Make it a habit to turn gas tanks off when not in use.

3. Familiarize yourself with your place of work and residence.

Identify relatively strong parts of the building like locations near elevator shafts, near columns and beams, and sturdy tables where you can take refuge during an earthquake.

Learn to use fire extinguishers, first aid kits, alarms, and emergency exits. These should be accessible/conveniently located, and prominently marked.

4. Prepare and maintain an earthquake survival kit consisting of a battery-powered radio, flashlight, first aid kit, potable water, candies, ready-to-eat food, sturdy rope, water-resistant matches, whistle, dust mask, clothes, and blanket.

5. Conduct and participate in regular earthquake drills.

During

1. Stay calm.

2. If you are inside a structurally sound building, stay there.

Protect your body from falling debris by staying in stronger parts of the building like near columns and beams, outside the elevator shaft, or by getting under a sturdy desk or table. Remember Duck, Cover, and Hold (Duck and Cover under a table, Hold on to the legs).

Stay away from glass windows, shelves, cabinets and other heavy objects.

Beware of falling objects. Be alert and keep your eyes open.

3. If you are outside, move to an open area.

Get away from power lines, posts, walls and other structures that may fall or collapse.

Stay away from buildings with glass panes.

If you are on a mountain or near a steep hill slope, move away from steep escarpments which may be affected by landslide.

4. When driving a vehicle, pull to the side of the road and stop. Do not attempt to cross bridges or overpasses, which may have been damaged.

5. If you are near the shore and you feel an earthquake strong enough to make standing difficult, it is always safest to assume that a tsunami might be triggered. Move quickly to higher ground.

After

1. Once the shaking stops, take the fastest and safest way out of the building. Do not rush to the exit. Get out calmly in an orderly manner. Bring your emergency kit. Do not use elevators; use the stairs. Go to an open area or designated evacuation area. Check yourself and others for injuries.

2. Unless you need emergency help:

Do not use your telephone to call relatives and friends. Disaster prevention authorities may need the line for emergency communication. Do not use your car and drive around areas of damage, as rescue and relief operations need the roads for mobility.

3. Help reduce the number of casualties from the earthquake.

Don't enter partially damaged buildings; strong aftershocks may cause these to collapse.

Gather information and disaster prevention instructions from battery-operated radios.

Obeey public safety precaution.

4. Check your surroundings.

Clean up chemical spills, toxic and flammable materials to avoid any chain of unwanted events.

Check for fire and, if any, have it controlled.

Check your water and electrical lines for defects; if any damage is suspected, turn the system off through the main valve or switch.

5. If you must evacuate your residence, leave a message stating where you are going.

What is a tsunami?

A tsunami is a series of small to giant sea waves commonly generated by under-the-sea earthquakes and whose heights may reach more than 5 meters. They are erroneously called tidal waves and are sometimes mistakenly associated with storm surges. Tsunamis can occur when the earth is shallow-seated and strong enough to displace parts of the seabed and disturb the mass of water over it.

Locally-generated tsunamis

The coastal areas in the Philippines, especially those facing the Pacific Ocean, South China Sea, Sulu Sea, and Celebes Sea, can be affected by tsunamis that may be generated by local earthquakes.

In August 17, 1976, a 7.9-magnitude earthquake hit the Moro Gulf and produced tsunamis that devastated the southwest coast of Mindanao and left more than 3,000 people dead, with at least 1,000 people missing. More than 8,000 people were injured and approximately 12,000 families were rendered homeless by more than 5-meter high waves.

The November 15, 1994 Mindoro Earthquake also generated tsunamis that left 78 casualties.

These tsunamis occurred within a very short time, with a first wave reaching the shoreline nearest the epicenter, 2 to 5 minutes after the main earthquake. These tsunamis were both locally-generated. There will not be enough time for warning in case of locally-generated tsunamis.

Far-field tsunamis

Tsunamis may also be generated from distant locations, such as those coming from

other countries bordering the Pacific Ocean like Chile, Alaska in the USA, and Japan (far field tsunamis). The tsunami of May 2, 1960 that was generated by a strong earthquake from Chile killed 61 in Hilo, Hawaii, while 20 people were reportedly killed in the Philippines. Travel times for tsunamis generated in distant locations are longer (1 to 24 hours) and will generally give enough time for warning from the Pacific Tsunami Warning Center (PTWC) and Northwest Pacific Tsunami Advisory Center (NWPTAC).

Natural signs of a local tsunami

1. You would feel the ground or earth shaking (earthquake).
2. Unusual sea level changes occur. There is sudden retreat or rise of sea water.
3. Rumbling sound of approaching waves is audible.

Tsunami Safety and Preparedness

Do not stay in low-lying coastal areas after a strong earthquake. Move to higher grounds immediately.

If unusual sea conditions like rapid lowering of the sea level are observed, immediately move toward high ground.

Never go down to the beach to watch for a tsunami. When you can see the wave, you are too close to escape it.

During the sudden retreat of sea level, interesting sights are often revealed. Fish may be stranded on dry land thereby attracting people to collect them. Sandbars and coral flats may also be exposed.

These scenes tend to attract people to flock to the shoreline, thereby increasing the number of people at risk.

Stay out of danger areas until an all clear bulletin is issued by a competent authority. A tsunami is not a single wave but a series of waves.

VOLCANIC ERUPTION

Why does the Philippines have many volcanoes?

The Philippines sits on a unique tectonic setting ideal to volcano formation. The archipelago is surrounded by subducting plates, as manifested by the trenches that are related to volcano formation.

Classification of volcanoes

1. Active volcanoes

Erupted within historical times (within the last 600 years). Accounts of these eruptions were documented by man;

Erupted within the last 10,000 years based on analyses of datable materials.

2. Potentially active volcanoes

Are morphologically young-looking but with no historical records of eruption.

3. Inactive volcanoes

Have no record of eruptions;

Have a physical form that is being changed by agents of weathering and erosion via the formation of deep and long gullies.

The six most active volcanoes in the Philippines

Volcano
Province
No. of Historical
Last Eruption

Eruptions

Mayon
Albay
49
2009
Taal
Batangas
33
1977
Bulusan
Sorsogon
17
2010-2011
Kanlaon
Negros Oriental
26
2006
Pinatubo
Boundaries
3
1992

of Pampanga,

Tarlac

and Zambales

Hibok-hibok
Camiguin
5
1948
Preparedness Measures

The first step toward safeguarding life, properties, and investments from volcanic hazards is understanding how and why they occur and know what areas are likely to be affected by these.

Comply strictly with PHIVOLCS' prohibition against human settlement in permanent danger zones. Always be on alert and listen to advisories of not venturing into identified restricted areas during volcanic unrest.

Identify, evaluate, and characterize areas vulnerable to hazards and implement appropriate land use plan and zonation.

Heed warnings and evacuation orders in times of volcanic unrest.

What to do before, during and after an ash fall

Before

1. When there is an impending eruption that may cause ashfall, listen to the radio for updates.
2. Prepare a gas mask or clean cloth for covering the nose, candles, matches, flashlights and extra batteries. Put these things in an accessible place.

During

1. Stay calm. Cover your nose and mouth with a damp, clean cloth or gas mask.
2. Listen to the radio for updates or developments regarding the volcanic eruption.
3. Keep your pets in their shelter or inside the house to keep them from inhaling ash.
4. Close all windows and doors of your house.
5. If you are driving a vehicle, pull to the side of the road and stop if there is heavy ashfall.
6. If you are outdoors, look for shelter and wear glasses to protect your eyes. Avoid using contact lenses.

After

1. As soon as the ashfall tapers, scrape off the ashfall that has accumulated on rooftops to prevent the collapse of houses. After removing the ash, clean the roof and gutter with water.
2. Wash all fruits and vegetables thoroughly before eating.
3. Use a vacuum cleaner or shake ash loose from the furniture before dusting them. Cover your nose and mouth while cleaning.
4. To remove ash from glass windows and doors, use a water hose before washing them with soap and lukewarm water.
5. Shake loose ash from plants before watering them.
6. Use powder detergent in washing clothes contaminated with ash.
7. Collect the ashes and put them in an area far from water drainage to avoid clogging.

TROPICAL CYCLONE

A tropical cyclone or -bagyo? is an intense low-pressure system with minimum sustained winds of 35 kilometers per hour (kph). A yearly average

of 20 tropical cyclones enter the Philippine Area of Responsibility (PAR) and 90% of these affect the country. Hazards due to tropical cyclones are strong winds with heavy rainfall that can cause widespread flooding/flashfloods, storm surges, landslides and mudflows.

Classification of Tropical Cyclones

Tropical cyclones are classified according to their strength and are determined by the speed of the maximum sustained winds near the center.

Tropical depression

-

35 kph to 63 kph

Tropical storm

-

64 kph to 117 kph

Typhoon

-

more than 118 kph

Supertyphoon

-

more than 220 kph

Structure of a Tropical Cyclone

The primary components of a Tropical Cyclone:

A strong tropical cyclone consists of the following:

Eye - is the center of the tropical cyclones and is normally circular in shape with generally cloud-free skies. The wind is light and calm or there are relatively very light winds and no rain. It ranges in size from 10 to 100 kilometers in diameter.

Eye wall- is the ring where very high winds and rains are at the heaviest. The highest winds are on the forward right side of the wall of the storm. If the storm is heading west, the highest winds will be on the northern side of the storm. The heaviest damage occurs when the tropical cyclone's eyewall passes over land.

Rainbands - the spiral rainbands of showers and thunderstorms surround the eye. High wind gusts and heavy downpour often occur in individual rainbands, with relatively calm weather between bands. It covers an area of several square kilometers.

PUBLIC STORM WARNING SIGNALS

Public Storm Warning Signal #1

METEOROLOGICAL CONDITIONS:

A tropical cyclone will affect the locality.

Winds of 30-60 kph may be expected within at least 36 hours or intermittent rains may be expected also within 36 hours. (When the tropical cyclone develops very close to the locality, a shorter lead time of the occurrence of the winds will be specified in the warning bulletin.) IMPACT OF THE WINDS:

Twigs and branches of small trees may be broken.

Some banana plants may be tilted or downed.

Some houses of very light materials (nipa and cogon) may be partially unroofed.

Unless this warning signal is upgraded during the entire existence of the tropical cyclone, only very light or no damage at all may be sustained by the exposed communities.

Rice crop, however, may suffer significant damage when it is in its flowering stage.

PRECAUTIONARY MEASURES:

When the tropical cyclone is strong or is intensifying and moving closer, this signal may be upgraded to the next level.

The waves on coastal waters may gradually develop and become bigger and higher.

The people are advised to listen to the latest severe weather bulletin issued by PAGASA every six hours. In the meantime, business may be carried on as usual except when floods occur.

Disaster preparedness is activated to alert status.

Public Storm Warning Signal #2

METEOROLOGICAL CONDITIONS:

A tropical cyclone will affect the locality.

Winds of greater than 60 kph and up to 100 kph may be expected in at least 24 hours.

IMPACT OF THE WINDS:

Some coconut trees may be tilted with a few others broken.

A few big trees may be uprooted.

Many banana plants may be downed.

Rice and corn may be adversely affected.

A large number of nipa and cogon houses may be partially or totally unroofed.

Some old galvanized iron roofing materials may be peeled off.

In general, the winds may bring light to moderate damage to the exposed communities.

PRECAUTIONARY MEASURES:

The sea and coastal waters are dangerous to small sea crafts.

Special attention should be given to the latest position, the direction and speed of movement, and the intensity of the storm as it may intensify and move toward the locality.

The general public, especially people travelling by sea and air, are cautioned to avoid unnecessary risks.

Outdoor activities of children should be postponed.

Secure properties before the signal is upgraded.

Disaster preparedness agencies/organizations are in action to alert their communities.

Public Storm Warning Signal #3

METEOROLOGICAL CONDITIONS:

A tropical cyclone will affect the locality.

Winds of greater than 100 kph up to 185 kph may be expected in at least 18 hours.

IMPACT OF THE WINDS:

Many coconut trees may be broken or destroyed.

Almost all banana plants may be downed and a large number of trees may be uprooted.

Rice and corn crops may suffer heavy losses.

Majority of all nipa and cogon houses may be unroofed or destroyed and there may be considerable damage to structures of light to medium construction.

There may be widespread disruption of electrical power and communication services.

In general, moderate to heavy damage may be experienced, particularly in the agricultural and industrial sectors.

PRECAUTIONARY MEASURES:

The disturbance is dangerous to the communities threatened/affected.

The sea and coastal waters will be very dangerous to all sea crafts.

Travel is very risky, especially by sea and air.

People are advised to seek shelter in strong buildings, evacuate low-lying areas, and stay away from the coasts and river banks.

Watch out for the passage of the -eye? of the typhoon, indicated by a sudden occurrence of fair weather immediately after very bad weather, with very strong winds coming generally from the north.

When the -eye? of the typhoon hits the community, do not venture away from a safe shelter, because after one to two hours the worst weather will resume with very strong winds coming from the south.

Classes in all levels should be suspended and children should stay in the safety of strong buildings.

Disaster preparedness and response agencies/organizations are in action with appropriate response to actual emergency. Public Storm Warning Signal #4

METEOROLOGICAL CONDITIONS:

A very intense typhoon will affect the locality.

Very strong winds of more than 185 kph may be expected in at least 12 hours.

IMPACT OF THE WINDS:

Coconut plantations may suffer extensive damage.

Many large trees may be uprooted.

Rice and corn plantations may suffer severe losses.

Most residential and institutional buildings of mixed construction may be severely damaged.

Electrical power distribution and communication services may be severely disrupted.

Damage to affected communities may be very heavy. PRECAUTIONARY MEASURES:

The situation is potentially very destructive to the community.

All travel and outdoor activities should be cancelled.

Evacuation to safer shelters should have been completed since it may be too late in this situation.

With PSWS #4, the locality is very likely to be hit directly by the eye of the typhoon. As the eye of the typhoon approaches, the weather will continuously worsen with the winds increasing to their strongest, coming generally from the north. Then, a sudden improvement of the weather with light winds (a lull) will be experienced. This means that the eye of the typhoon is over the locality. This improved weather may last for one to two hours depending on the diameter of the eye and the speed of movement. As the eye moves out of the locality, the worst weather experienced before the lull

will suddenly commence. Thereafter, very strong winds will come generally from the south.

The disaster management councils concerned and other disaster response organizations are now fully responding to emergencies and in full readiness to immediately respond to a possible calamity.

What to do in case of typhoons

Stay indoors and keep calm.

Monitor TV and radio reports.

Secure your home.

Trim trees near dwellings.

Keep roads clear for emergency vehicles.

Go to the nearest designated evacuation center if your house is in a flood-prone area.

Have a flashlight and radio handy, with fresh batteries.

Stock up on food, potable water, kerosene, batteries, and first-aid supplies.

In case of flooding, turn off the main sources of electricity, gas and water in your home.

Stack furniture above the expected flood level. Keep appliances, valuables, chemicals, toxic substances, and garbage beyond the reach of floodwaters.

Avoid low-lying areas, riverbanks, creeks and coastal areas, slopes, cliffs, and foothills. Rain can trigger landslides, rockslides or mudslides.

Avoid wading through flooded areas. Do not attempt to cross flowing streams.

Do not operate any electrical equipment during a flood.

Do not use gas or electrical appliances that have been flooded.

Flood

FLOOD is due to water overflowing from streams and other bodies of water, as well as by the accumulation of rainwater by drainage. Floods can cause severe damages to lives and properties.

Flood/Flooding/Inundation - is a great flow of water that rises and spreads over the land when a volume of water from lakes/streams/rivers exceeds its carrying capacity and escapes from its usual boundaries.
Types of Floods

Flooding can be classified based on the following criteria:

Based on location or place of occurrence

- River flooding is caused by the overflowing of

the river when run-off exceeds the capacity of the channels and flooding the adjacent low-lying areas called the floodplain. This may last a few hours or many days depending on the intensity, amount and the distribution of the rainfall.

- Coastal flooding may occur due to storm surges, high tide and tsunamis (waves produced by earthquakes/volcanoes at sea). Storm surge occurs when strong onshore winds push the water inland and cause a rise in sea level, and flooding the low-lying coastal areas. Worst cases could occur when it coincides with the occurrence of high tide.

- Urban flooding occurs in an area where roads are usually paved. During rainy episodes, water cannot infiltrate the ground and is normally retained in the surface. This type of flooding is often associated with the limited capacity of the sewerage system to drain the heavy rains that are falling.

Based on duration of occurrence

- Flash Flooding is caused by a very short period of unusually heavy rainfall or cloudburst over a relatively small drainage area. Flash floods carry highly destructive flood waves and are most common in mountainous areas or in steep places that have streams flowing through narrow canyons.

- Sheet Flooding is caused by comparatively shallow water flowing over a wide area and is very common in floodplains that are normally flat. Sheet flooding may also result when water from a river channel with insufficient carrying capacity over-tops its bank, inundating the adjacent areas.

Causes of Flooding

Floods are due to the complex combination of weather, climatic and human activities:

1. Heavy, continuous rain that does not cease, or ceases only briefly.
2. Meandering river system
3. Heavy siltation of the river system which could decrease the carrying capacity of the river
4. Dam spilling/over-topping of dikes and levees
5. Unusually high coastal and estuarine waters due to storm surges
6. Damming - Sudden discharge of floodwater with accompanying debris materials, including trees, rocks, soil, etc.
7. To a certain extent, astronomically influenced phenomena such as high tides coinciding with the occurrence of heavy rainfall

Aggravating factor:

- Presence of informal settlers
- Increasing population
- Increasing level of urbanization

- Altering the ecological system in a river basin, which could have an impact on the hydrology of the watershed

- Denudation of forest and watershed areas

If it has been raining hard for several hours, or steadily raining for several days, be alert to the possibility of a flood. Floods happen as the

ground becomes saturated.

Pre-Flood Preparation

- Avoid building your house on the floodplains and riversides.

- Keep drainage areas free from debris accumulation.

- Ensure that drainage facilities are functioning properly.

- Plant trees and protect the forests, especially in the mountains, to diminish floods.

- Protect the rivers. Do not throw garbage because it will make them shallow, destroying their draining capabilities.

- Do not connect septic tanks to the drainage canals and rivers.

- Periodic desilting, dredging, widening and deepening of riverbeds must be conducted.

Before the Flood

- Be informed of the daily weather conditions and forecasts from PAGASA.

- Be aware of how often and to what extent your location is likely to be flooded.

- Know the flood warning system and evacuation plan of your community and make sure your family knows them.

- Identify an evacuation area for the family and livestock.

- Assign specific tasks and responsibilities to family members

according to an evacuation plan.

- Keep a stock of food, which requires no or little cooking and refrigeration, good for at least three days.

- Keep a battery-operated radio and flashlight, cell phones, emergency cooking equipment, candles, matches, and a first aid kit handy in case of emergency.

- Securely anchor weak houses.

When Warned of Flood

- Watch for rapidly rising flood waters.

- Listen to your radio for emergency and possible evacuation orders or instructions.

- Store drinking water in containers as water service may be interrupted.

- Move household belongings to upper levels.

- Transfer livestock to higher ground.

- If advised to evacuate, do so. Don't panic. Move to a safe area before access is cut off by floodwaters.

- Turn off the main electricity switch and gas valve. Disconnect electrical appliances. Do not touch electrical equipment if you are wet or standing in water. Lock your house before evacuating.

During the Flood

- Stay indoors.

- Do not attempt to cross rivers or flowing streams where water is above the knee.

- Do not go swimming or boating in swollen rivers.

- Do not drive through flooded areas or streets. Avoid needless trips.

- Stay away from downed power lines.

- Be alert for gas leaks. Use a flashlight to inspect for damage.

- Report flooded areas to the Local Disaster Risk Reduction and Management Councils.

After the Flood

- Re-enter your house with caution using a flashlight, not lanterns or torches. Flammables and dangerous animals like snakes may be inside.

- Be alert for fire hazards like broken electric wires.

Do not eat food and drink water until they have been checked for contamination.

Report broken utility lines (electricity, water, gas and telephone) to appropriate agencies/authorities.

Do not turn on the main switch or use appliances and other equipment until they have been checked by a competent electrician.

Consult health authorities for immunization requirements.

Do not go sight-seeing in disaster areas. Your presence might hamper rescue and other emergency operations.

Storm Surge

A STORM SURGE is a sudden rise in sea level above normal, causing big waves as a tropical cyclone approaches the coast. The highest water level rise or the peak of a storm surge usually happens on the same time there is a passage of a typhoon across a coastline. The wall of water can engulf low-lying coastal communities and can also bring destruction to natural and man-made structures, especially if this coincides with the occurrence of high tide.

The stronger the tropical cyclone, the lower the atmospheric pressure; the shallower the coast, the higher the surge will be. A storm surge can sweep the coastline, and can extend several kilometers inland.

What to do

Make plans for evacuating members of your family and yourself to higher ground before a storm surge takes place.

Stay off the beach when a weather disturbance is approaching or exists in your community.

Listen to the PAGASA Public Forecast/Warnings that are aired regularly. The storm surge warning is incorporated in the Domestic Bulletins and reads like this: Residents along low-lying coastal areas are advised to seek higher ground and to take the necessary precautions against flooding due to big waves from the sea.

As soon as possible, evacuate to higher ground and stay away from beaches when a typhoon approaches your community.

Thunderstorm

A THUNDERSTORM is a violent local atmospheric disturbance accompanied by lightning, thunder, and heavy rain, often with strong gusts of wind and sometimes by hail. The typical thunderstorm is 5 kilometers in diameter and lasts an average of 30 minutes, but the more severe ones may last much longer. When compared with tropical cyclones, thunderstorms affect relatively small areas but all thunderstorms are dangerous.

Hazards associated with thunderstorms

1. Lightning

Lightning is a visible electrical discharge due to high

voltage differences that build up within the cloud, or between the rain cloud and the ground below.

The temperature of a lightning bolt is five times hotter than the surface of the sun. The sudden lightning creates sound waves in the air that we hear as thunder. Lightning can cause death and injuries and can

spark fires.

2. Heavy Rainfall

Excessive rains can produce flashfloods/floods that can transform small creeks into raging floodwaters, sweeping over structures along their path.

3. Tornado/Waterspout

The more severe hazard brought about by a thunderstorm is a tornado/waterspout. The amount of destruction depends largely on the strength of the wind. Most injuries and deaths result from flying debris.

4. Hail

Hailstones are ice crystals falling from a thunderstorm cloud. Since the air temperature in the country is normally high, hailstones that reach the ground are usually the size of a grain of corn, which is seldom of sufficient size to cause damage to properties. However, hail can destroy crops.

What are the signs of an impending thunderstorm?

- Towering cumulonimbus clouds
- Darkening skies
- Flashes of lightning and gusty winds
- Sound of thunder

Who are at risk from a thunderstorm?

People who are outdoors (especially under or near tall trees, in isolated sheds or near hilltops) are at risk from lightning. Also in danger are people in automobiles when flash flooding occurs near them.

How to prevent damages due to lightning

If you are outdoors, seek shelter from lightning. Go inside the house, enclosed car or large building, if you can.

If you can't find shelter, avoid the solitary tree or the tallest objects in the area. Lightning tends to strike tall objects.

Stay away from anything that conducts electricity.

Don't use any plug-in electrical appliances such as tele- phones, computers, or television sets.

Stay in your automobile if you are travelling.

When outside, don't use metal objects.

Get out of water; it conducts electricity.

When you feel an electrical charge-your hair stands on end or your skin tingles-lightning may be about to strike you. Don't drop to the ground. Sit down instead, but with only your feet touching the ground and positioned apart from each other while you cover your head with your hands or hold your knees.

Remember, if you can hear thunder, you are close enough to be struck by lightning!

Tornado

A TORNADO is described as a violently rotating column of air extending from a thunderstorm to the ground. Tornadoes come in many sizes but are typically in the form of a visible condensation funnel whose narrow end touches the earth and is often encircled by a cloud of debris.

A tornado acts like a giant vacuum cleaner, sucking and carrying objects-such as trees, structures, or debris-along on its path and jumping across the ground in a narrow

erratic movement.

It can strike any time of the day but does so more frequently in the afternoon-when the temperature is at its highest-and in the evening. It may also develop

when there are tropical cyclones.

Much of the damage comes from its extreme winds. Wind speeds are estimated to exceed 450 km/hr and may last for a few minutes. Whenever a tornado strikes, it may leave marks of destruction. Tornadoes are often obscured by rain or dust.

There have been reported occurrences of -buhawi,? the local term for tornado, in the Philippines. One occurred in San Miguel, Bulacan, and damaged at least 30 houses in 2007.

Signs of an impending tornado

A funnel-shaped column appears from the base of a cumulonimbus cloud.

The weather feels warm, humid, uncomfortable and calm.

Lightning flashes and thunder cracks.

A tornado reaching the ground produces a roaring or buzz-ing sound.

What is a waterspout?

A tornado that occurs or moves over a body of water is called a waterspout. On the average, a waterspout is less intense and less violent, but the larger ones can completely destroy a small sea craft.

Cases of waterspout coming ashore with the intensity of a tornado have been reported. In December 1979, one struck Gingoog City in Misamis Oriental, and rendered 10,000 families homeless.

How can I keep myself safe?

If at work, home or in school, keep calm, take cover in any interior hallway, the lowest place in a building, or under sturdy furniture-and brace yourself. Stay away from windows.

Secure outdoor objects (like garbage cans, garden tools, toys, signs, garden sets) if time permits, to prevent them from becoming missiles during the tornado's passage.

Lie down in any depression or crawl into culverts when a tornado is about to strike as it is considerably better than remaining upright.

If driving a vehicle, stop and abandon it and seek shelter outside in the nearest depression, ditch or ravine (but be aware of flash flooding).

Tornadoes are generally short-lived. They come and go quickly. There may not be time for a warning.

Be alert for the sudden appearance of violent winds, rain, hail or a funnel-shaped cloud in your locality.

Landslide

A LANDSLIDE is the downward and outward movement of slope materials composed of rocks and/or soil. This event is generally triggered by prolonged heavy rainfall or by earthquakes. A

number of factors contribute to the occurrence of landslides. These include the high degree of weathering and fracturing of rocks, steep configuration of slope, deforestation, and the overloading of slope surfaces following the construction of

heavy structures.

What to do before:

Coordinate with the Mines and Geosciences Bureau (MGB) regarding the landslide susceptibility of the area concerned.

Consult a professional for advice on engineering measures

most appropriate for the site for landslide mitigation or prevention.

Reinforce the foundation and walls of structures when needed.

Develop a land use plan, ensuring that site developments are compatible with the conditions and limitations of the natural environment

Promote public awareness and involvement on landslide mitigation.

Develop a community preparedness and evacuation plan.

Maintain a list of contact numbers during emergencies.

What to do during:

Evacuate the community immediately if warned of an impending landslide

Get out of the house/building as soon as possible when rumbling sounds are heard from upstream or when trembling of the ground is felt. Run across a slope, not downwards.

Stay away from the path of landslide debris, or seek refuge behind a sturdy tree or structure.

What to do after:

Ask proper authorities to thoroughly examine damaged structures and utilities before using these.

Stay away from the landslide area. There may be danger of additional landslides.

Check with caution injured and trapped persons within the landslide area. Direct rescuers to their locations.

Listen to local radio or television stations for the latest emergency information.

Seek the advice of an expert for evaluating landslide hazards or designing corrective techniques to reduce landslide risk.

Extreme Climatic Variability

CLIMATE change is the direct impact of global warming with observed increase in global mean temperatures. These rising temperatures will cause

changes in weather patterns that may result to increased frequency and intensity of extreme weather events such as droughts and floods affecting the most vulnerable areas of the country and the whole region as well. Longer drought periods and more intense rainfall will greatly affect water resources. Other impacts include an increasing number of intense tropical cyclones, storm surges, killer heat waves and the emergence of more weather-related diseases, massive species extinction, ecosystem collapse, agricultural failure due to changes in rainfall patterns, higher temperature and saltwater intrusion, and rising sea levels which will eventually inundate coastal towns and cities around the world.

Periodical Phenomenon of Climatic Variability

An extreme climate event associated with the El Nino Southern Oscillation (ENSO) influences the climate variability of the country. It is a global phenomenon that occurs in the equatorial Pacific Ocean, affecting the normal rainfall pattern of the different regions of the world, but impacts differ. ENSO has two phases: the warm phase referred to as El Nino and the cold phase referred to as La Nina.

La Nina is characterized by unusually cold surface ocean temperatures in the equatorial Pacific Ocean, as compared to El Nino, which is characterized by unusually warm surface ocean temperatures. The system may be described generally by an irregular cycle of alternating periods of warm (El Nino) and cold (La Nina) conditions, with a transition period (Neutral condition) between them. El Nino and La Nina occur on average of every 3 to 5 years. Generally, the impact of El Nino in the country is drier than normal rainfall condition, which

sometimes leads to drought, and La Nina to be associated with wetter than normal condition, causing devastating floods and landslides.

Heat Waves

Heat Waves occur when the daily maximum temperature of more than five consecutive days exceeds the average maximum temperature by 5-degree Celcius, the normal period

being 1961-1990 (World Meteorological Organization).

Heat can kill by pushing the human body temperature beyond its limits. Elderly people, young children, and those who are sick or overweight are more likely to become victims

of extreme heat. Men sweat more than women so they are more susceptible to heat illness and become more quickly dehydrated. Excessive heat that lasts for more than two days significantly increases heat-related illnesses. People living in urban areas may be at greater risk from the effects of a prolonged heat wave than people living in rural regions.

What to do before:

- Plan changes in your daily activities to avoid strenuous work during the warmest part of the day.

- Consult a physician when affected by extreme heat.

- Undergo training on first aid.

- Interview local physicians and paramedics about the dangers of sunburn, heat exhaustion, heat stroke, and other possible conditions caused by excessive heat.

What to do during:

- Wear loose-fitting, lightweight, light-colored clothing that will cover as much skin as possible.

- Protect your face and head by wearing a wide-brimmed hat.

- Drink plenty of water even if you do not feel thirsty to prevent dehydration.

- Avoid drinks with alcohol or caffeine, which can also cause dehydration.

- Take frequent breaks if you must work outdoors.

- Use a buddy system when working in extreme heat.

- Eat frequent small meals for easy digestion.

- Use salt tablets or table salt because it retains fluids.

Human-Induced Hazards

Structure Collapse

Structure collapse is often caused by engineering failures such as the under-designing of structural components, by corrosion attack, and by an aerodynamic resonance in structures.

What to do before:

- Conduct a general check on the condition of all houses and buildings

- Repair or rehabilitate structures to put them in good condition.

- Adopt the existing warning system for a building certified as unsafe for occupancy.

- Cordon off and mark buildings found unsafe with -off- limits?.

What to do during:

Vacate the house/building immediately.

Apply first aid, and in cases of injuries or casualties, bring the victims to the nearest hospital for medical treatment. What to do after:

Secure the area.

Evaluate and assess the damaged structure through the assistance of engineers.

Recommend for approval the demolition of condemned buildings, subject to recommendation by proper authorities. Fire

Fire is composed of three elements-heat, fuel, and oxygen, which when combined, will result in a chemical reaction called burning. The leading cause of death in a fire, by a three-to-one ratio over burns, is asphyxiation (choking sensation). Inhaling carbon monoxide can cause loss of consciousness or death within minutes. The heat from a hostile fire exceeds anything to which a person is normally exposed. A fully developed room fire has temperatures over 1, 100 degrees Fahrenheit.

Fire generates a black, impenetrable smoke that blocks vision and stings the eyes. It is impossible to navigate through such smoke, so fire drill participants should practice evacuating houses/ buildings by at least two routes.